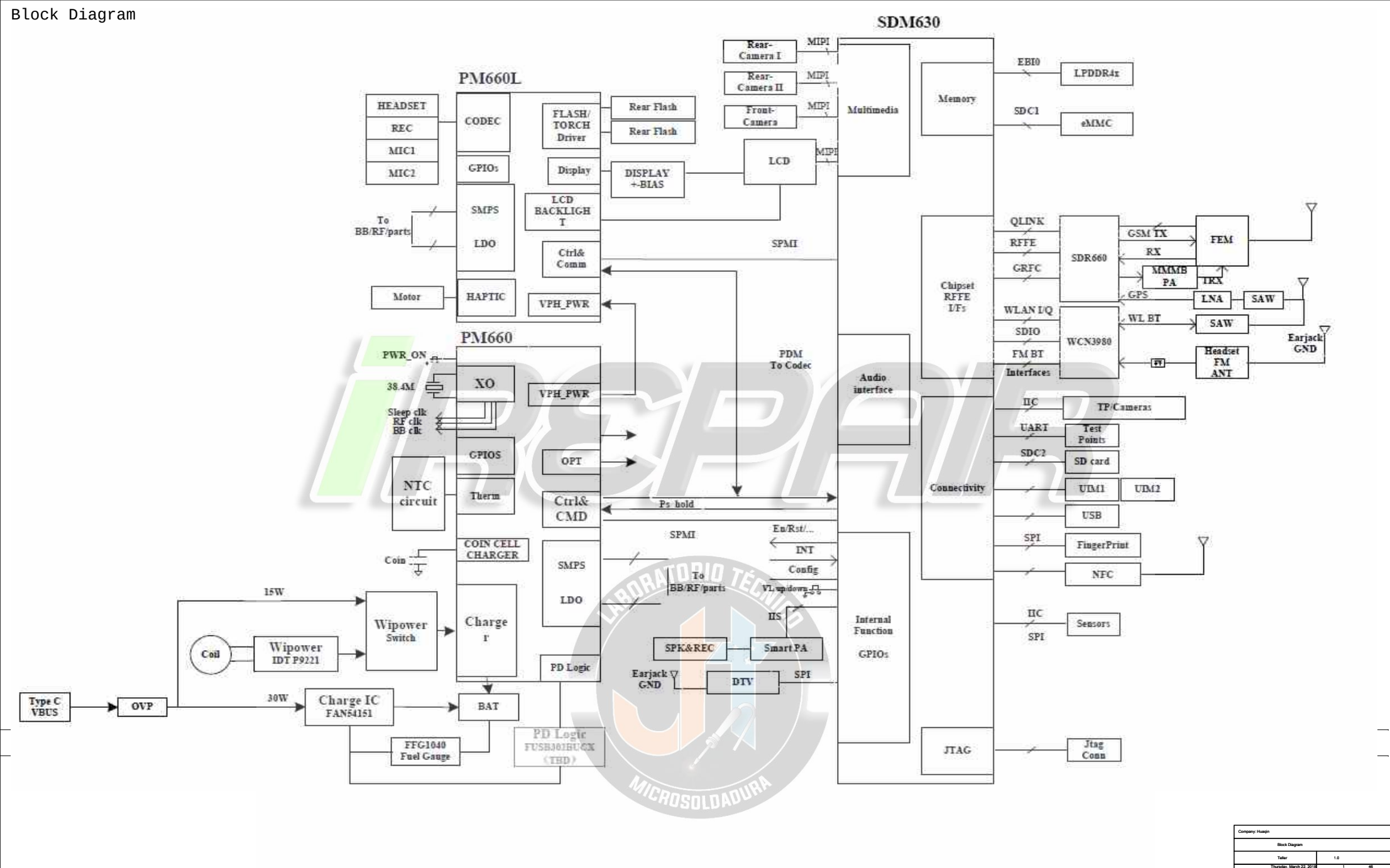




I2C Configuration Table

I2C	Function	I2C Slave Address(7-bit mode)
BLSP2_I2C	eMCP DCDC	FAN53526 I2C Adress:0x60 (Write:0xC0 Read:0xC1) PM3003A I2C Adress:0x60 (Write:0xC0 Read:0xC1) RT5738 I2C Adress:0x53 (Write:0xA6 Read:0xA7)
	2nd Charger IC	BQ25970-Default NC I2C Adress:0x?? (Write:0x?? Read:0x??)
	Smart PA	CS35L35 I2C Adress:0x40 (ADR=0 Write:0x80 Read:0x81)
	HPUE	TPS61280 I2C Adress:0x75 (Write:0xEA Read:0xEB)
	Sar Sensor	CY8C4024FNI-S412 I2C Adress:0x?? (Write:0x?? Read:0x??)
BLSP3_I2C	TP	GT917D I2C Adress:0x5D (Write:0xBA Read:0xBB) FT5446 I2C Adress:0x38 (Write:0x70 Read:0x71)
BLSP6_I2C	NFC	PN553 I2C Adress:0x28 (Write:0x50 Read:0x51)
CCI_I2C0	Dedicated Cam I2C0	S5K2L7SA-Rear Main CAM I2C Adress:0x2D (Write:0x5A Read:0x5B)
CCI_I2C1	Dedicated Cam I2C1	S5K4H7YX-Front Cam I2C Adress:0x10 (Write:0x20 Read:0x21) OV5675-Rear Slave Cam I2C Adress:0x36 (Write:0x6C Read:0x6D)
LPI_I2C	ALS&PS	TMD27253 I2C Adress:0x39 (Write:0x72 Read:0x73)
	eCompass	AK09918C I2C Adress:0x0C (Write:0x18 Read:0x19)

SPI Configuration Table

SPI	Function
BLSP1_SPI	FPS interface
BLSP4_SPI	DTV interface
LPI_SPI	A+G sensor interface

I2S Configuration Table

I2S	Function
I2S_2	Smart PA CS35L35 interface
I2S_1_MCLK	Smart PA CS35L35 mclk

Table of Content

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05.	SDM630 Control/UFS/SDC1/SDC2	28.	REAR_CAM/FLASH
06.	SDM630 LPDDR4	29.	FRONT_CAM/CAM_POWER
07.	SDM630 CSI/DSI	30.	SIM_SD_KEY_LINK_BATT
08.	SDM630 GPIO (Internal Codec)	31.	DTV
09.	SDM630 Power (Core)	32.	NFC
10.	SDM630 Power (Other, LP4X)	33.	PRIMARY_ANTENNA
11.	SDM630 Ground	34.	PA & PA THERMAL
12.	PM660 CLKs/Control_GND	35.	TRX_LB
13.	PM660 Buck Converter	36.	TRX_MHB
14.	PM660 LDO Circuits	37.	DIVERSITY_ANT
15.	PM660 Charger_FG	38.	DRX_MB_HB
16.	PM660L Control and Bob_Flash	39.	DRX_LB
17.	PM660L Buck and LDO	40.	SDR660_MSS
18.	PM660L CODEC	41.	SDR660_PWR
19.	PM660L Display	42.	SDR660_DA/PRX/DRX/FBRX
20.	eMCP	43.	QFE4101
21.	30W CHARGE	44.	WIFI_2.4G_5G & BT
22.	Wipower	45.	GPS
23.	Test Points	46.	WCN3980



GPIO MAP

SDM630 GPIO Configuration

GPIO_0	BLSP_SPI_MOSI_1	GPIO_42	NC	GPIO_84	UIM2_CLK
GPIO_1	BLSP_SPI_MISO_1	GPIO_43	NC	GPIO_85	UIM2_RESET
GPIO_2	BLSP_SPI_CS1_N_1	GPIO_44	NC	GPIO_86	UIM2_PRESENT
GPIO_3	BLSP_SPI_CLK_1	GPIO_45	NC	GPIO_87	UIM1_DATA
GPIO_4	BLSP_UART_TX_2	GPIO_46	CAM0_RST_N	GPIO_88	UIM1_CLK
GPIO_5	BLSP_UART_RX_2	GPIO_47	CAM1_RST_N	GPIO_89	UIM1_RESET
GPIO_6	BLSP_I2C_SDA_2	GPIO_48	CAM2_RST_N	GPIO_90	UIM1_PRESENT
GPIO_7	BLSP_I2C_SCL_2	GPIO_49	NC	GPIO_91	UIM_BATT_ALARM
GPIO_8	BLSP_SPI_MOSI_3	GPIO_50	CAM_AF_VDD_EN	GPIO_92	NC
GPIO_9	BLSP_SPI_MISO_3	GPIO_51	CAM_AVDD_EN	GPIO_93	NC
GPIO_10	BLSP_SPI_CS_N_3	GPIO_52	NC	GPIO_94	NC
GPIO_11	BLSP_SPI_CLK_3	GPIO_53	LCD0 RESET_N	GPIO_95	NC
GPIO_12	NC	GPIO_54	SD_CARD_DET_N	GPIO_96	WD0G_DISABLE
GPIO_13	KEYPAD_LED_PWM	GPIO_55	DP_EN_N	GPIO_97	NC
GPIO_14	BLSP_I2C_SDA_4	GPIO_56	USBC_ORIENTATION	GPIO_98	NC
GPIO_15	BLSP_I2C_SCL_4	GPIO_57	FORCED_USB_BOOT	GPIO_99	QLINK_REQUEST
GPIO_16	BLSP_UART_TX_5	GPIO_58	USB_PHY_PS	GPIO_100	QLINK_ENABLE
GPIO_17	BLSP_UART_RX_5	GPIO_59	MDP_VSYNC_P	GPIO_101	RFFE1_DATA - SDR660 debug
GPIO_18	BLSP_UART_CTS_N_5	GPIO_60	NC	GPIO_102	RFFE1_CLK - SDR660 debug
GPIO_19	BLSP_UART_RFR_N_5	GPIO_61	NC	GPIO_103	RFFE2_DATA - (41)QAT3533
GPIO_20	FP_SUB_RESET	GPIO_62	NC	GPIO_104	RFFE2_CLK - (41)QAT3533
GPIO_21	SMB_STAT	GPIO_63	NC	GPIO_105	NC
GPIO_22	BLSP_I2C_SDA_6	GPIO_64	NC	GPIO_106	NC
GPIO_23	BLSP_I2C_SCL_6	GPIO_65	NC	GPIO_107	RFFE4_DATA-(45)QSW8574,(48)RF1656SR
GPIO_24	CDC_SWR_CLK	GPIO_66	TS_RESET_N	GPIO_108	RFFE4_CLK -(45)QSW8574,(48)RF1656SR
GPIO_25	CDC_SWR_DATA	GPIO_67	TS_INT_N	GPIO_109	RFFE5_DATA-(52)QAT3514, (52)QAT3550,(47) QAT3522
GPIO_26	WSA_SPKR_SD_N_1	GPIO_68	ACC_GYRO_DATA_AVA_INT_N	GPIO_110	RFFE5_CLK-(52)QAT3514, (52)QAT3550,(47)QAT3522
GPIO_27	NC	GPIO_69	ACC_GYRO_MOT_INT_N	GPIO_111	RFFE6_DATA - (46)QET4101 ch0, (41)QPA4361
GPIO_28	NFC_IRQ	GPIO_70	MAG_INT_N	GPIO_112	RFFE6_CLK - (46)QET4101 ch0, (41)QPA4361
GPIO_29	NFC_EN	GPIO_71	ALSP_INT_N	GPIO_113	NC
GPIO_30	NFC_DWL_REQ	GPIO_72	FP_INT_N_1	LPI_GPIO_0	LPI_SPI_1_CS1_N
GPIO_31	NFC_ESE_PWR_REQ	GPIO_73	NC	LPI_GPIO_1	LPI_PWR_EN
GPIO_32	CAM_MCLK0	GPIO_74	NC	LPI_GPIO_2	LPI_I2C_3_SDA
GPIO_33	CAM_MCLK1	GPIO_75	NC	LPI_GPIO_3	LPI_I2C_3_SCL
GPIO_34	CAM_MCLK2	GPIO_76	NC	LPI_GPIO_4	LPI_SPI_2_CS_N
GPIO_35	NC	GPIO_77	AUDIO_USBC_EN2_N	LPI_GPIO_5	LPI_SPI_2_CLK
GPIO_36	CCI_I2C_SDA0	GPIO_78	WMSS_RESETN	LPI_GPIO_6	LPI_SPI_2_MOSI
GPIO_37	CCI_I2C_SCL0	GPIO_79	PWM_CNTRL	LPI_GPIO_7	LPI_SPI_2_MISO
GPIO_38	CCI_I2C_SDA1	GPIO_80	AUDIO_USBC_EN1	LPI_GPIO_8	LPI_SPI_1_CS_N
GPIO_39	CCI_I2C_SCL1	GPIO_81	MSS_LTE_COXM_TXD	LPI_GPIO_9	LPI_SPI_1_CLK
GPIO_40	NC	GPIO_82	MSS_LTE_COXM_RXD	LPI_GPIO_10	LPI_SPI_1_MOSI
GPIO_41	FL_STROBE_TRIG	GPIO_83	UIM2_DATA	LPI_GPIO_11	LPI_SPI_1_MISO

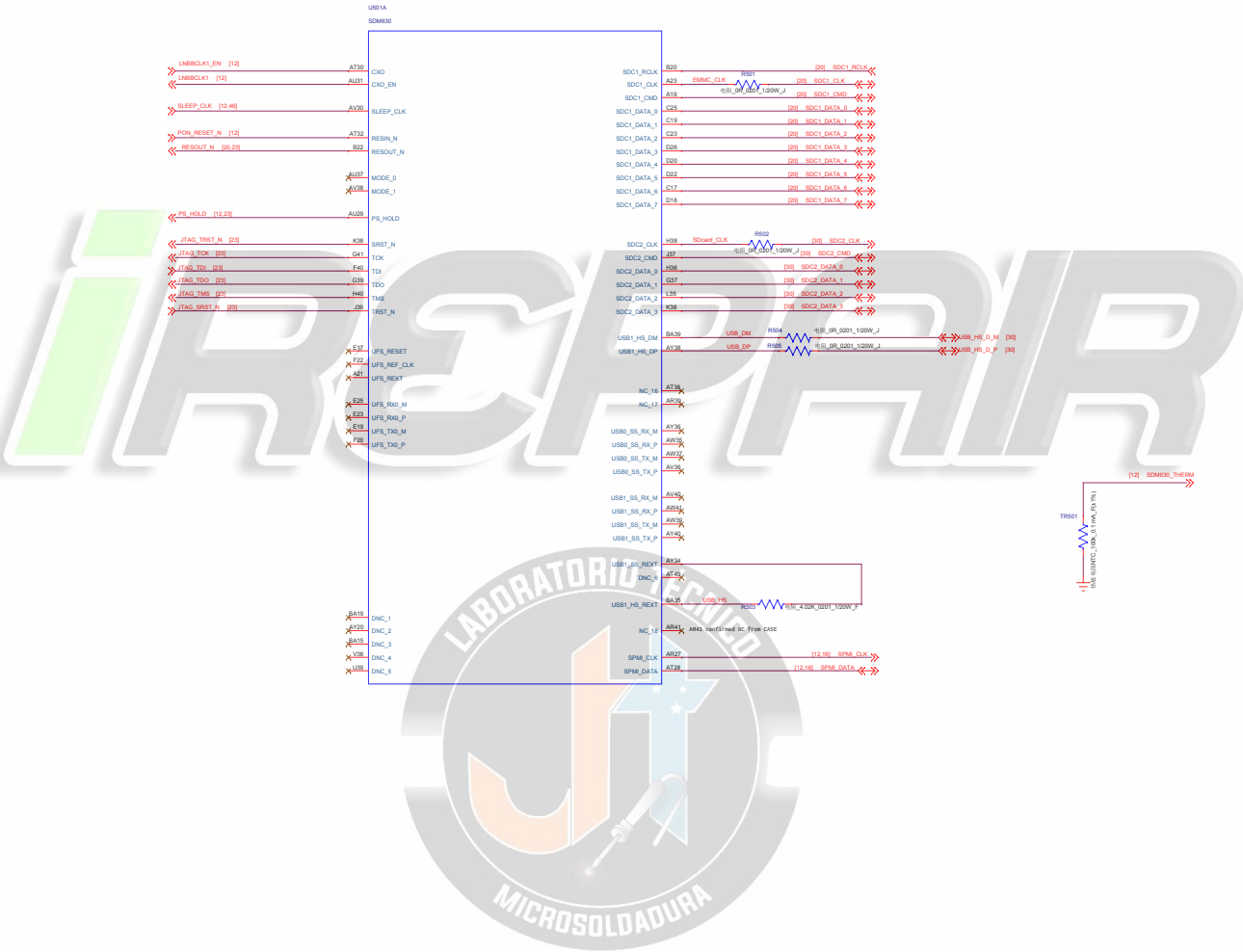
LPI_GPIO_12	LPI_UART_1_TX
LPI_GPIO_13	LPI_UART_1_RX
LPI_GPIO_14	NC
LPI_GPIO_15	NC
LPI_GPIO_16	NC
LPI_GPIO_17	NC
LPI_GPIO_18	CDC_PDM_CLK
LPI_GPIO_19	CDC_PDM_SYNC
LPI_GPIO_20	CDC_PDM_TX
LPI_GPIO_21	CDC_PDM_RX0
LPI_GPIO_22	CDC_PDM_RX0_COMP
LPI_GPIO_23	CDC_PDM_RX1
LPI_GPIO_24	CDC_PDM_RX1_COMP
LPI_GPIO_25	CDC_PDM_RX2
LPI_GPIO_26	CDC_DMIC_CLK1
LPI_GPIO_27	CDC_DMIC_DATA1
LPI_GPIO_28	CDC_DMIC_CLK2
LPI_GPIO_29	CDC_DMIC_DATA2
LPI_GPIO_30	LPI_QCA_SB_CLK
LPI_GPIO_31	LPI_QCA_SB_DATA0

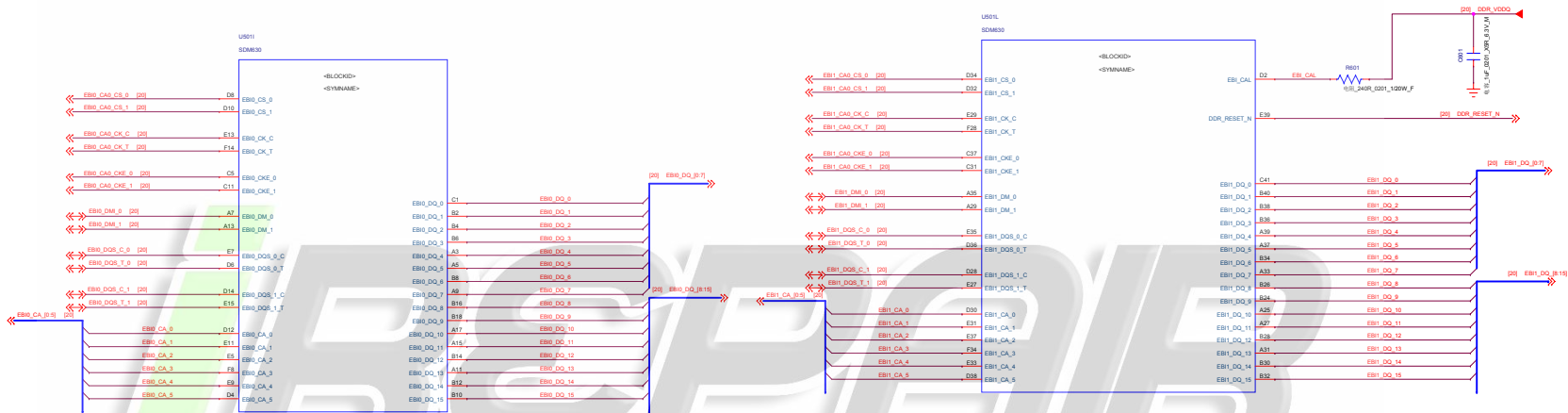
PM660 GPIO Configuration

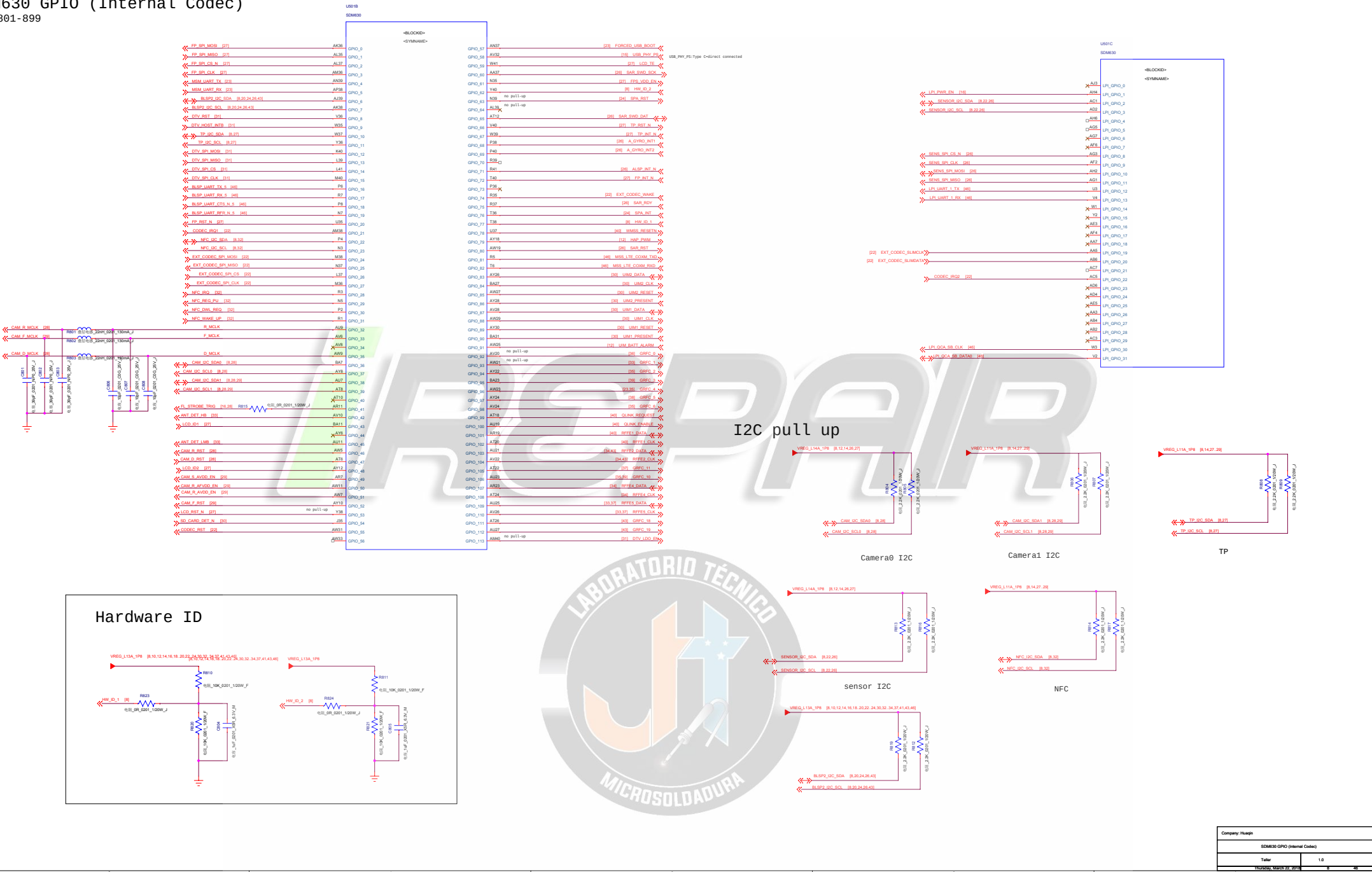
GPIO_1	OPTION1	GPIO_8	SLB
GPIO_2	DIV_CLK2	GPIO_9	uUSB_TYPEC
GPIO_3	NC	GPIO_10	WCSS_VCTRL
GPIO_4	NFC_CLK_REQ	GPIO_11	HOMEKEY_FP_PM_INT
GPIO_5	WLAN_SW_CTRL	GPIO_12	WIPWR_MODE
GPIO_6	SLP_CLK	GPIO_13	PM_A_GPIO_13
GPIO_7	UIM_BATT_ALARM		

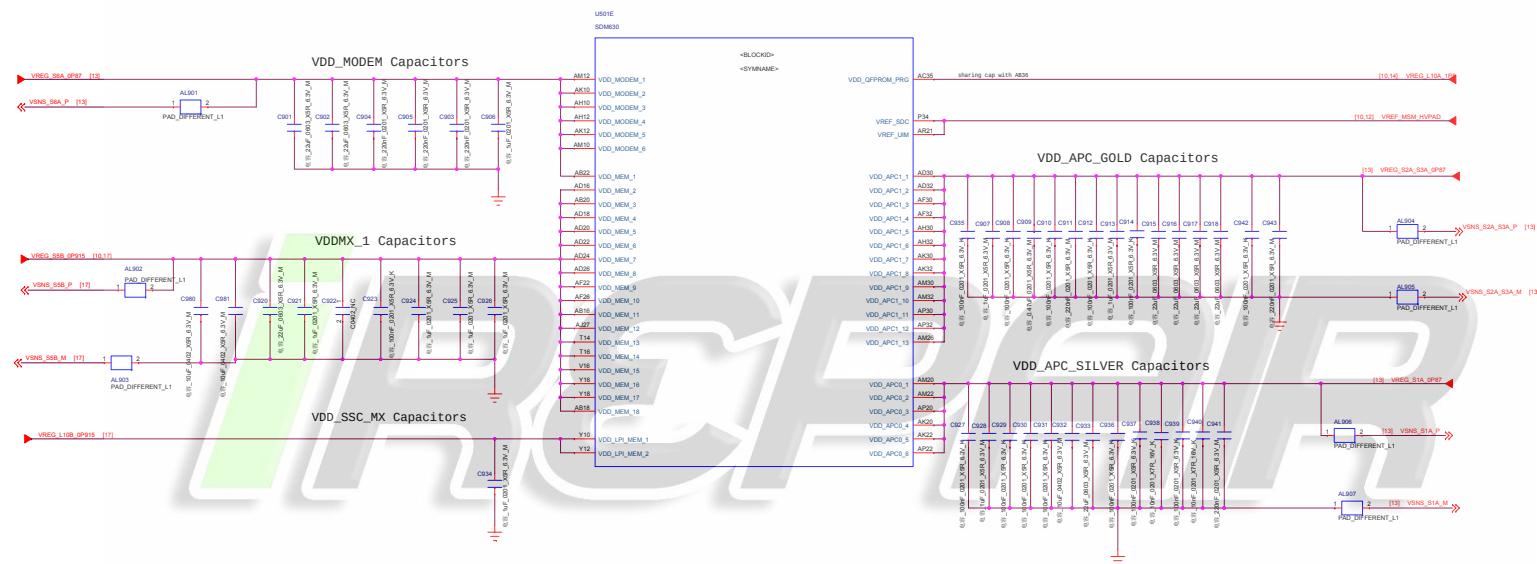
PM660A GPIO Configuration

GPIO_1	OPTION2	GPIO_8	SD_CARD_DET_N
GPIO_2	LPI_PWR_EN	GPIO_9	WCSS_VCTRL
GPIO_3	FRONT_CAM_DVDD_EN	GPIO_10	SLB
GPIO_4	REAR_CAM_DVDD_EN	GPIO_11	LP4x_CTRL
GPIO_5	NC	GPIO_12	LP4x_MODE
GPIO_6	FP_VDD_EN		
GPIO_7	KEY_VOL_UP_N		

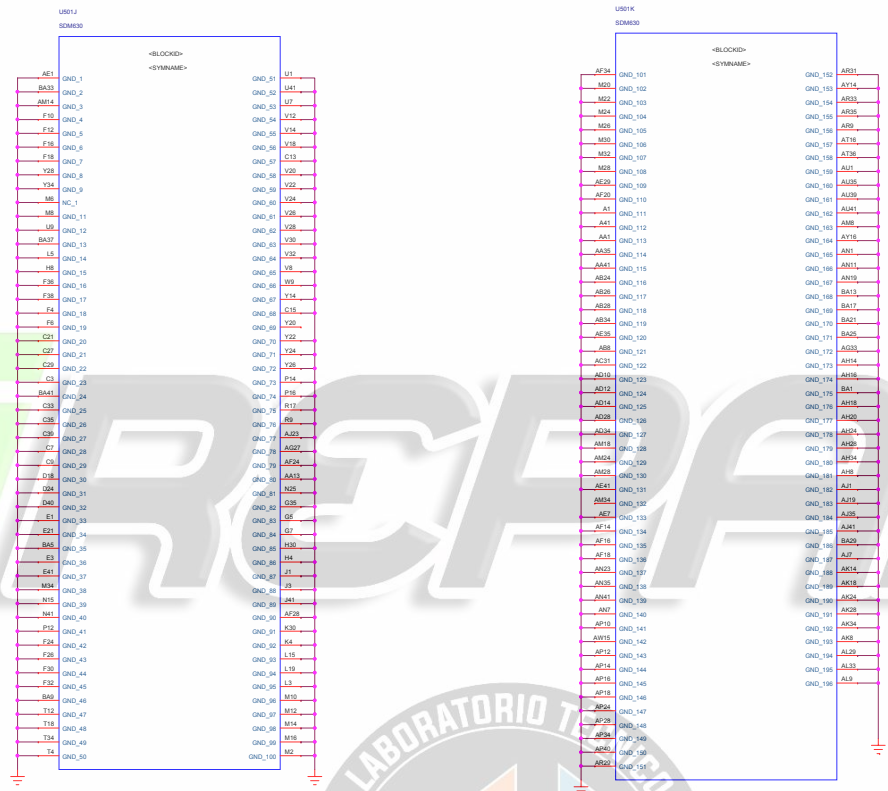




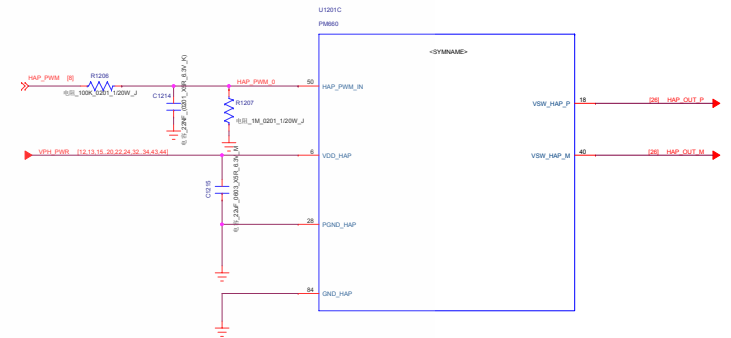
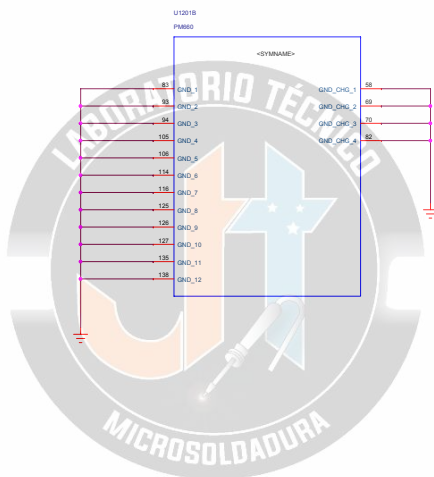
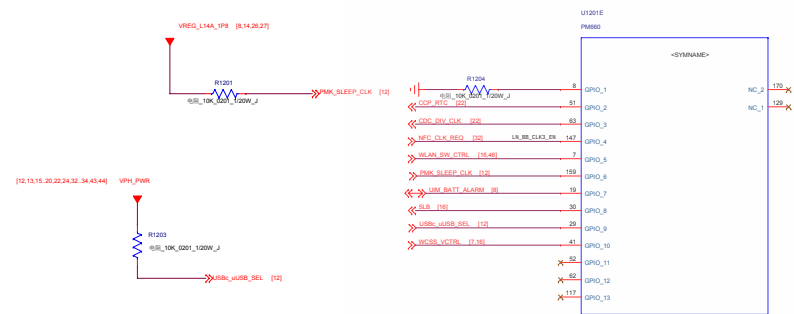
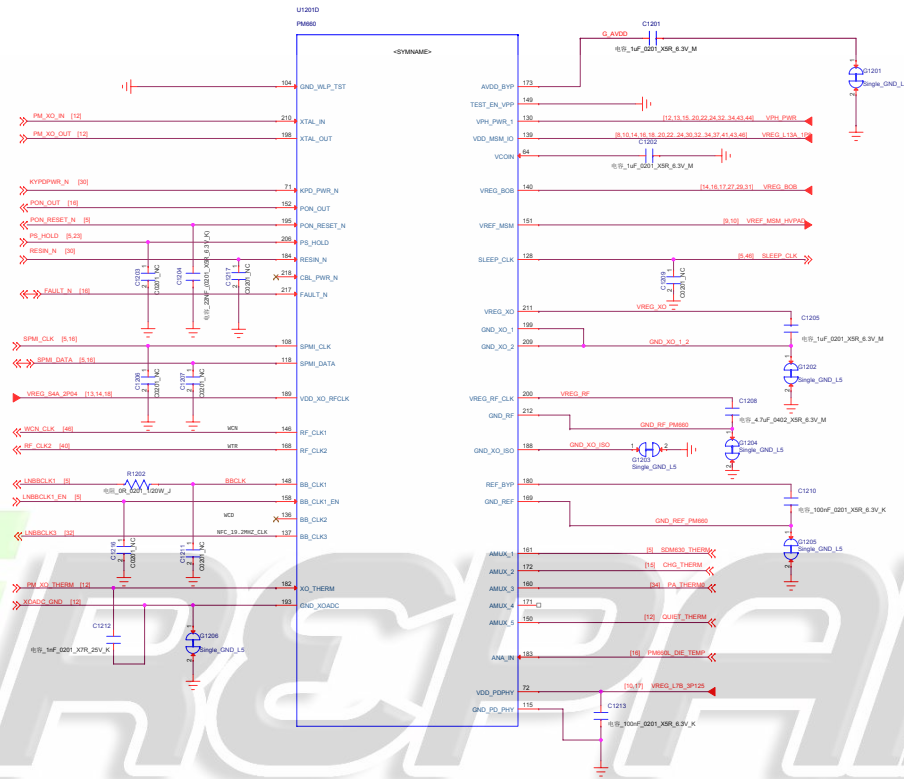


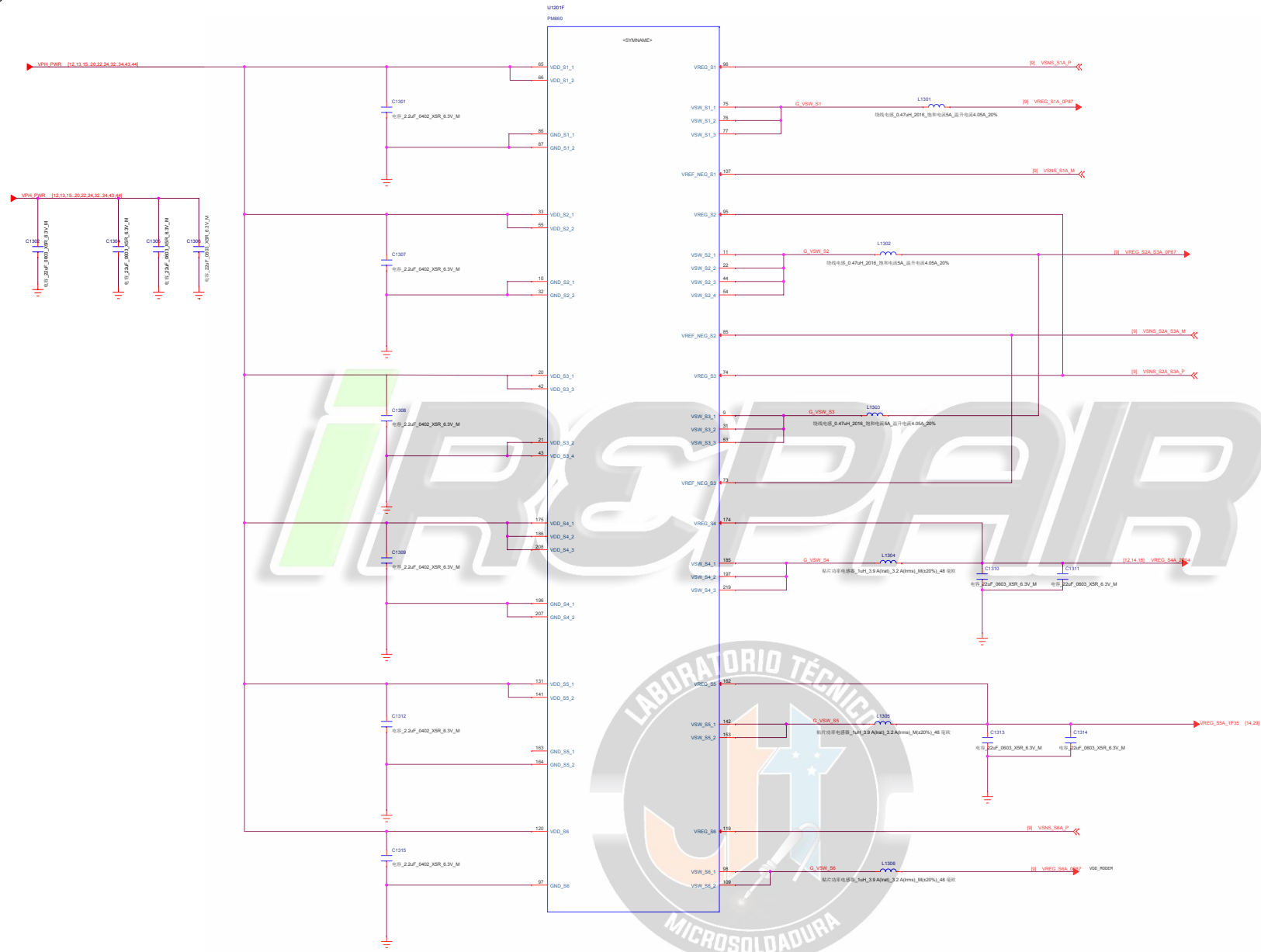


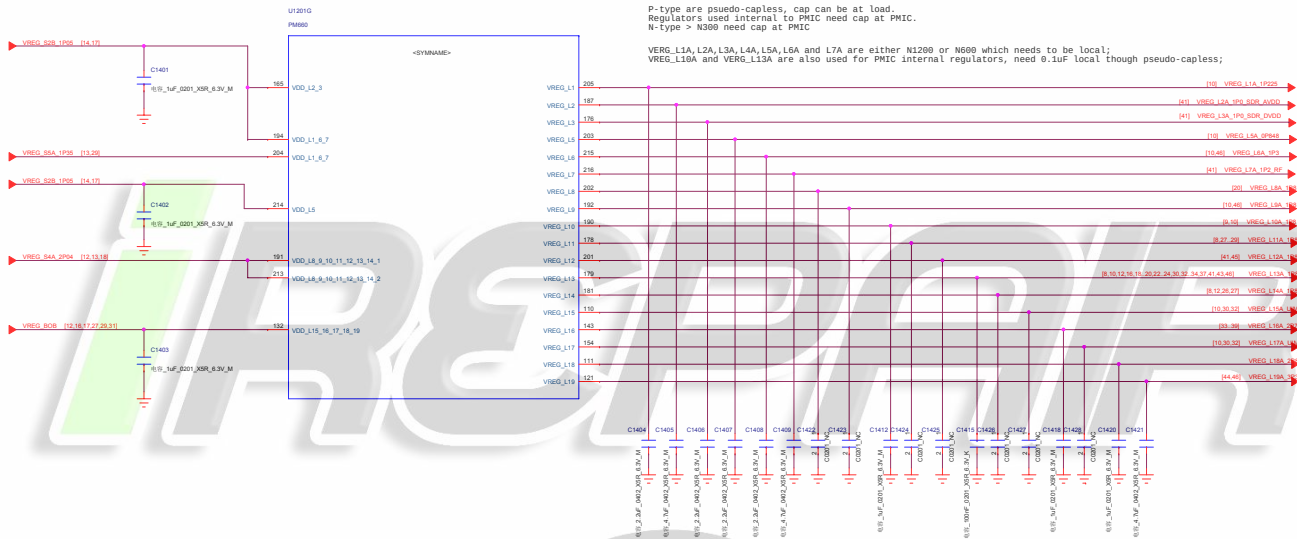
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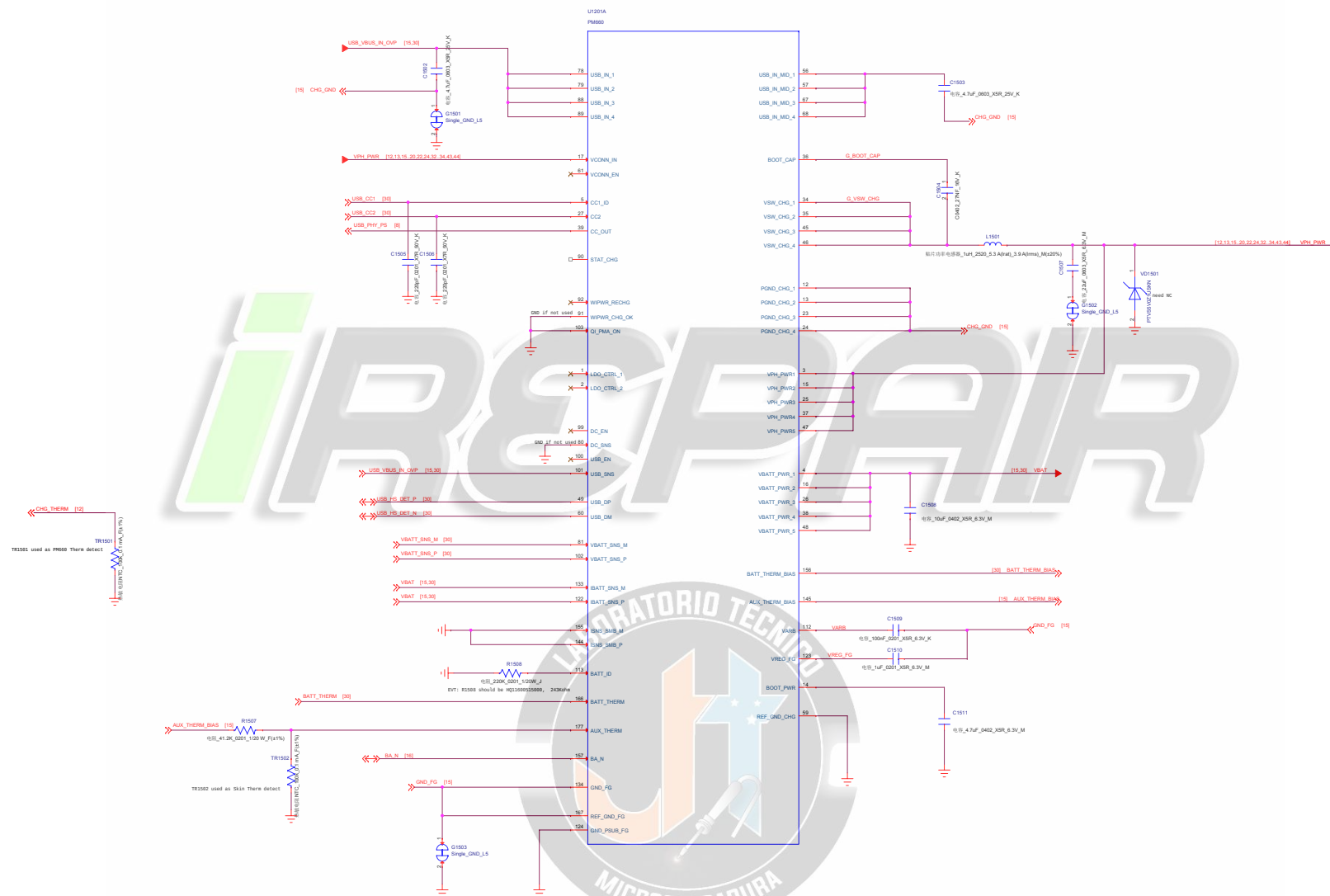


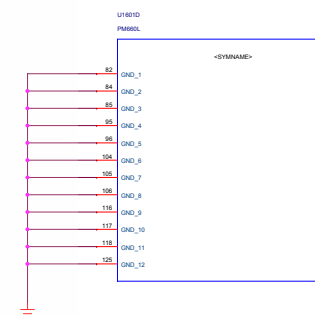
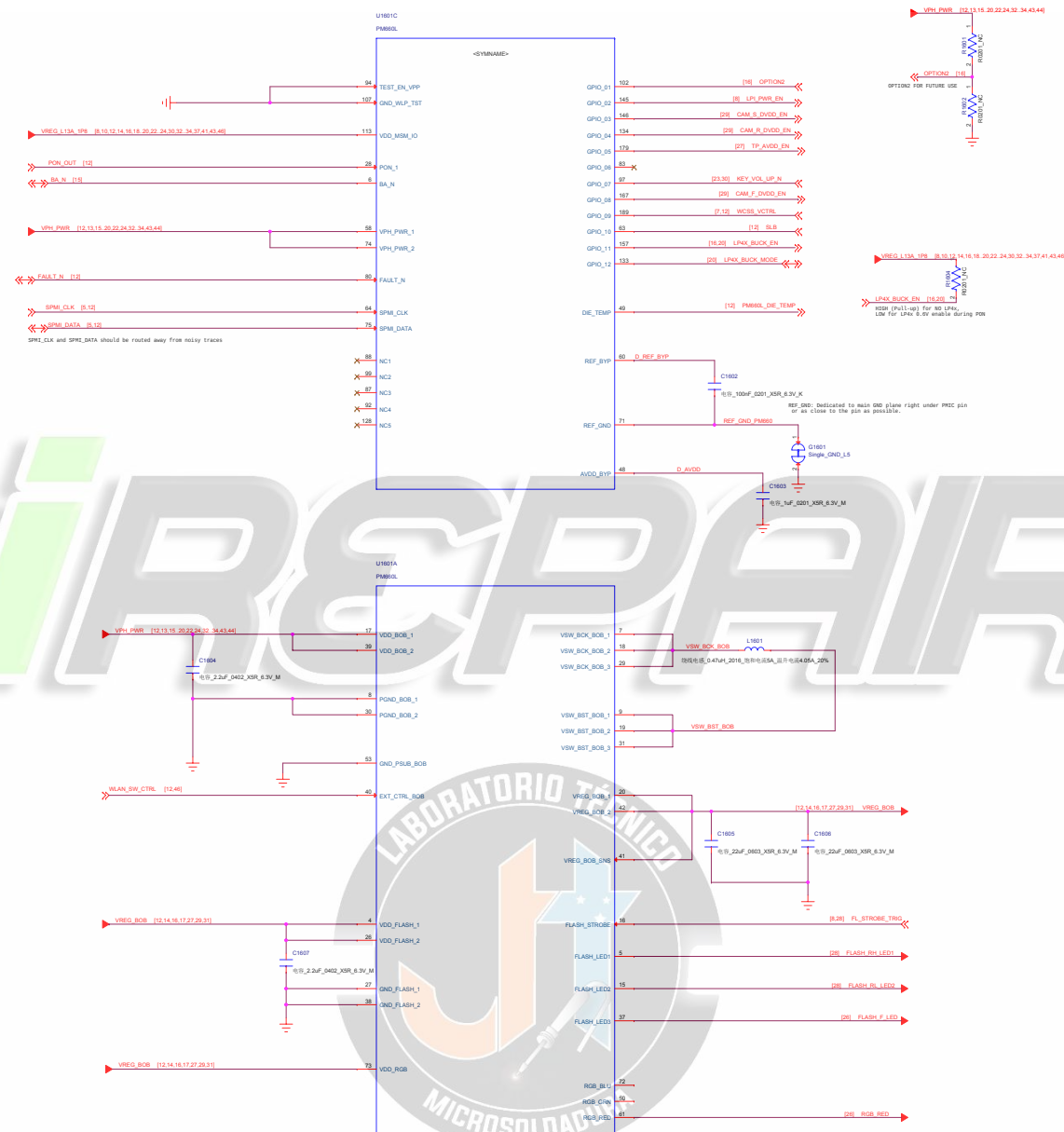
PM660 CLKs/Control_GND
REF 1201-1299

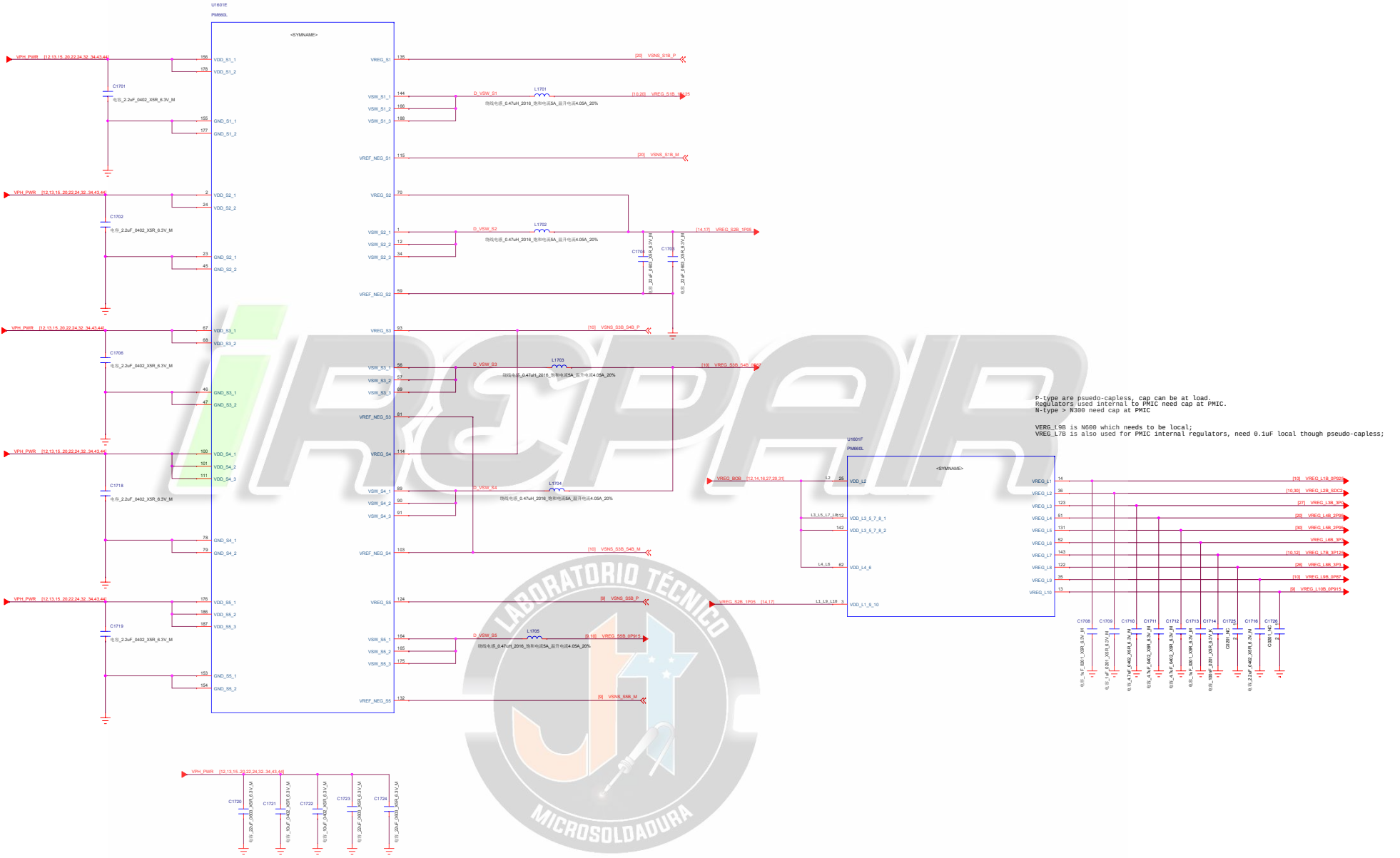






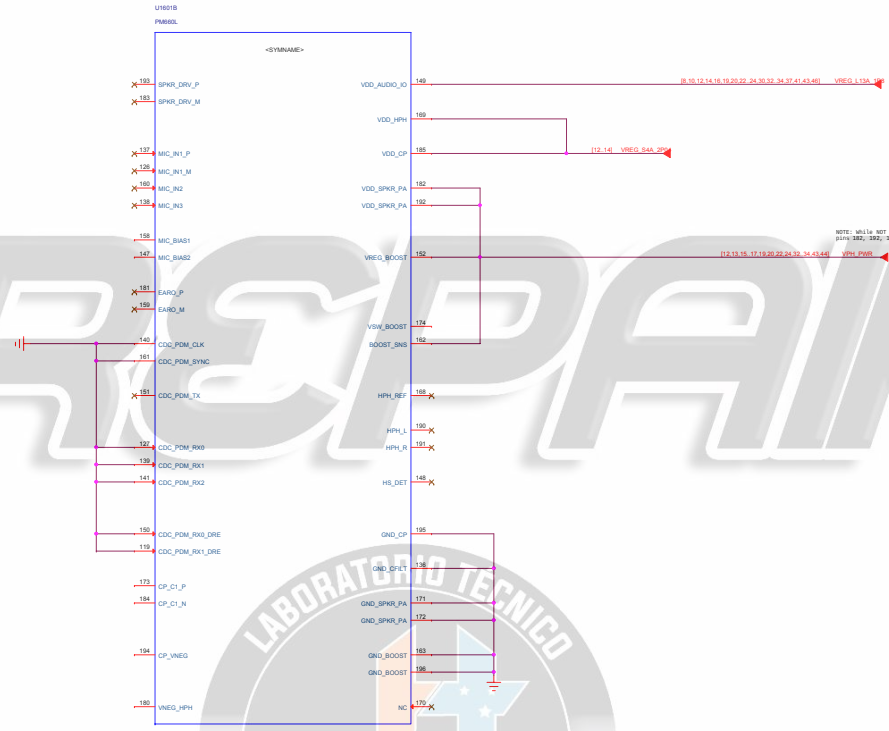
PM660 Charger_FG
REF 1501-1599

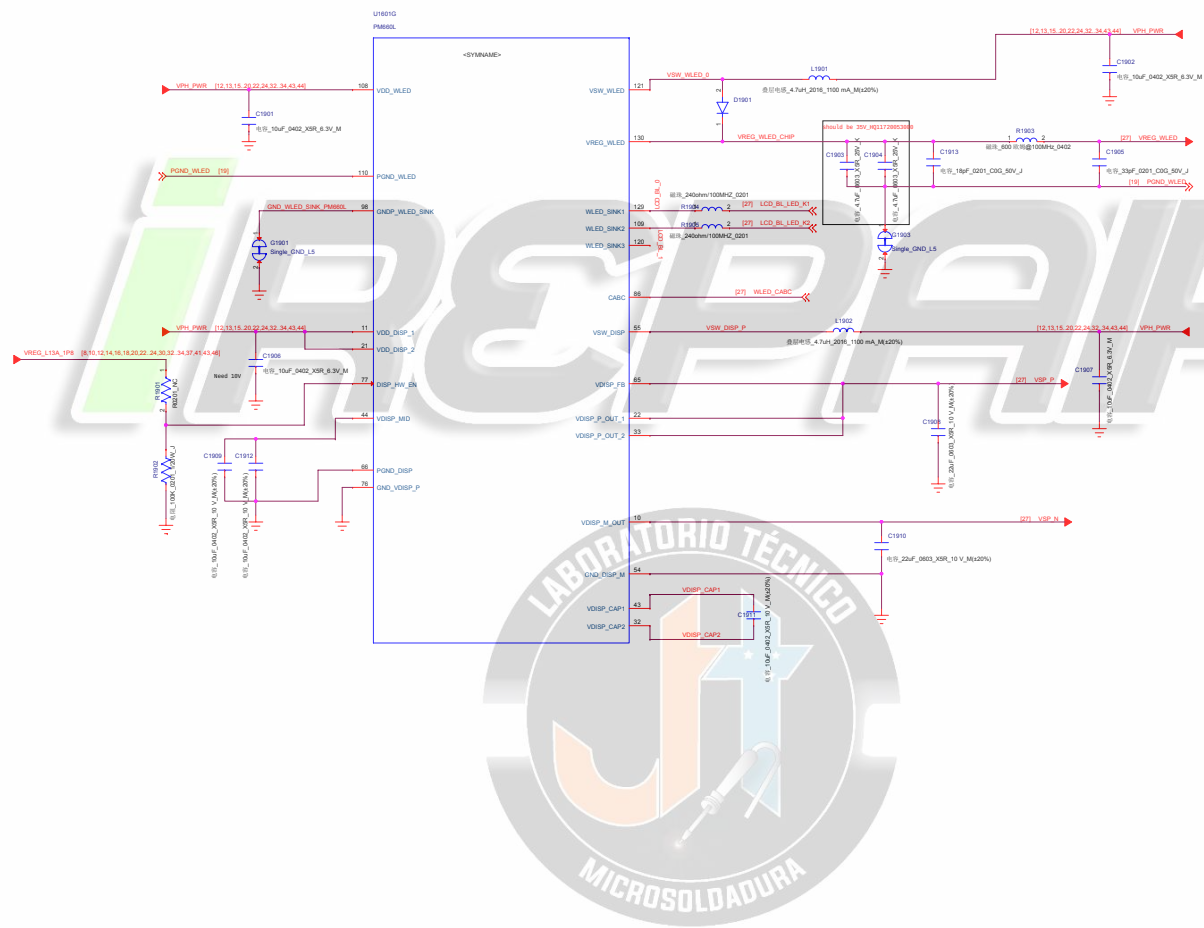


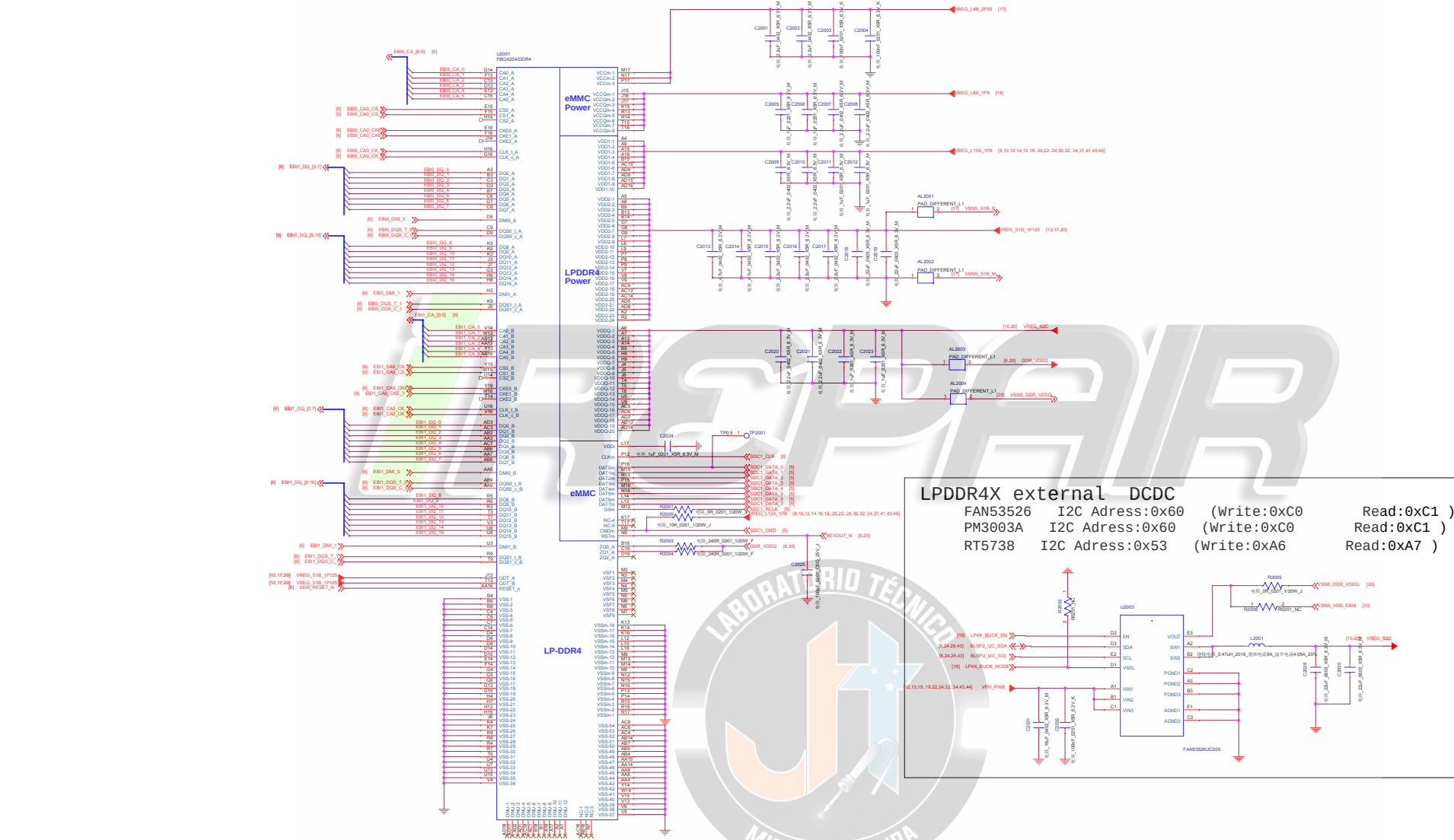


P-type are pseudo-capless, cap can be at load.
Regulators used internal to PMIC need cap at PMIC.
N-type > N300 need cap at PMIC.

VREG_198 is N600 which needs to be local;
VREG_178 is also used for PMIC internal regulators, need 0.1uF local though pseudo-capless;







LPDDR4X external DCDC

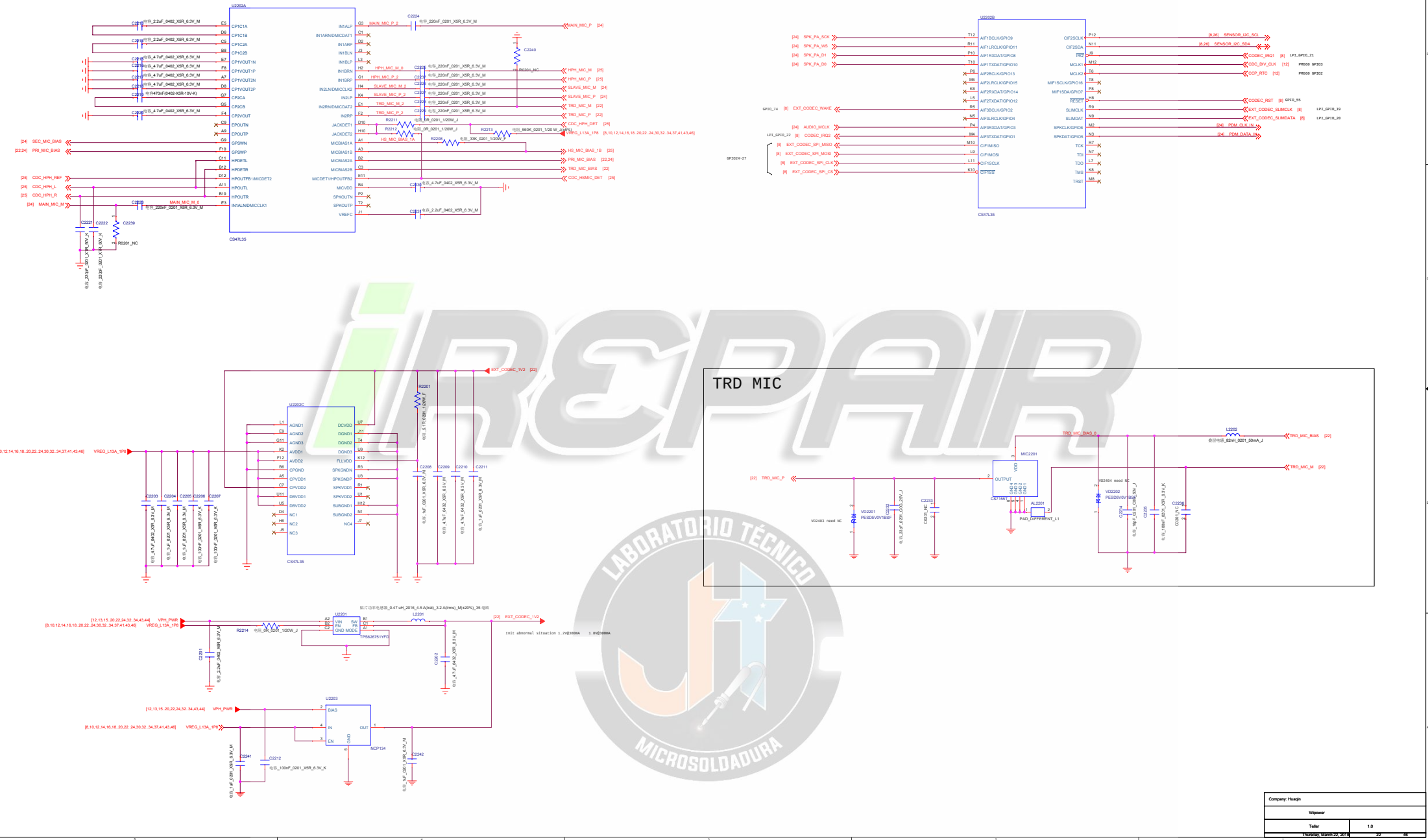
Device	I2C Address	Write	Read
FAN53526	0x60	0xC0	0xC1
PM3003A	0x60	0xC0	0xC1
RT5738	0x53	0xA6	0xA7

The circuit diagram shows the LPDDR4X external DCDC circuit. It includes the FAN53526, PM3003A, and RT5738 regulators. The regulators are connected to the LPDDR4X memory module via a series of capacitors and resistors. The diagram also shows the connection of the regulators to the power rails.

iREPAIR



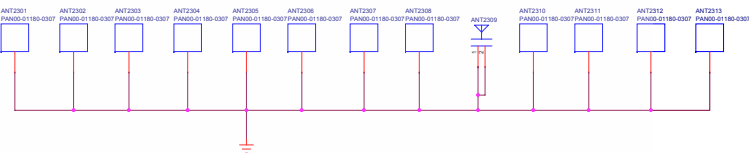
Company Name	
24W CHARGE	
Taller	1.0
Escriba el nombre de su taller	



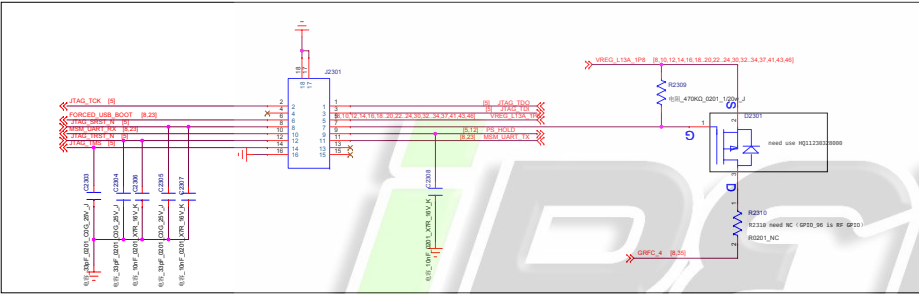
UART and Force USB boot



Grounding Springs



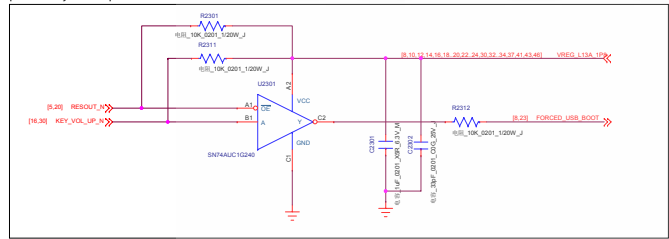
JTAG CONNECTOR



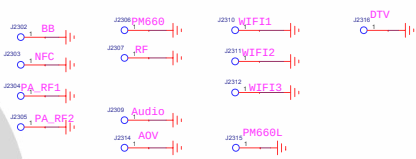
Cable clamp



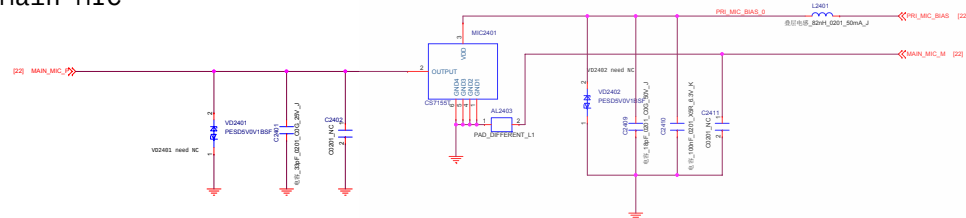
power key +volup to enter 9908 mode



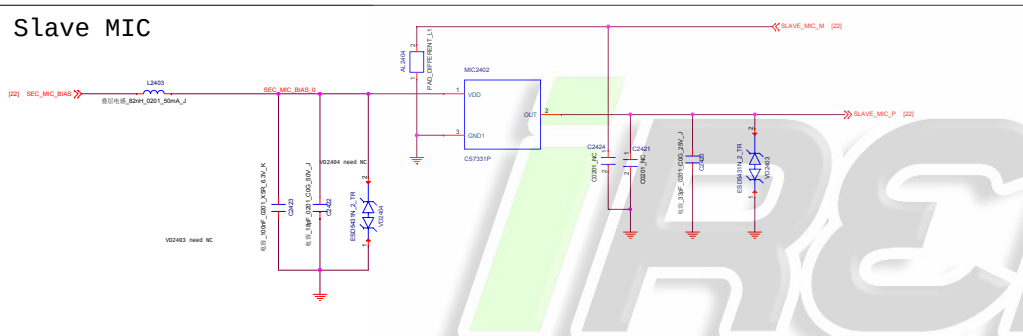
Shields



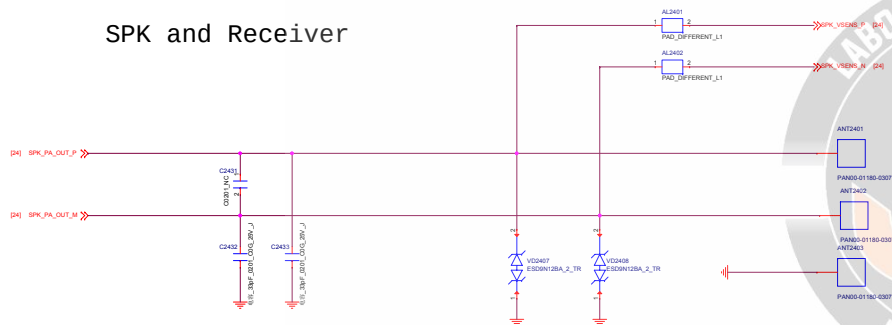
Main MIC



Slave MIC



SPK and Receiver



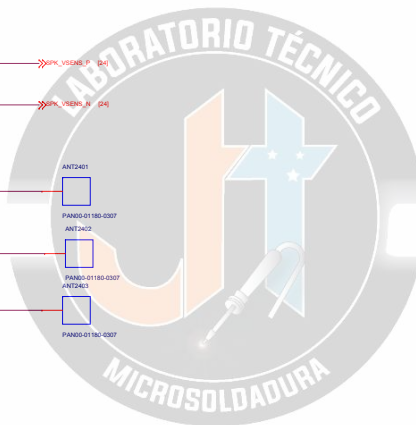
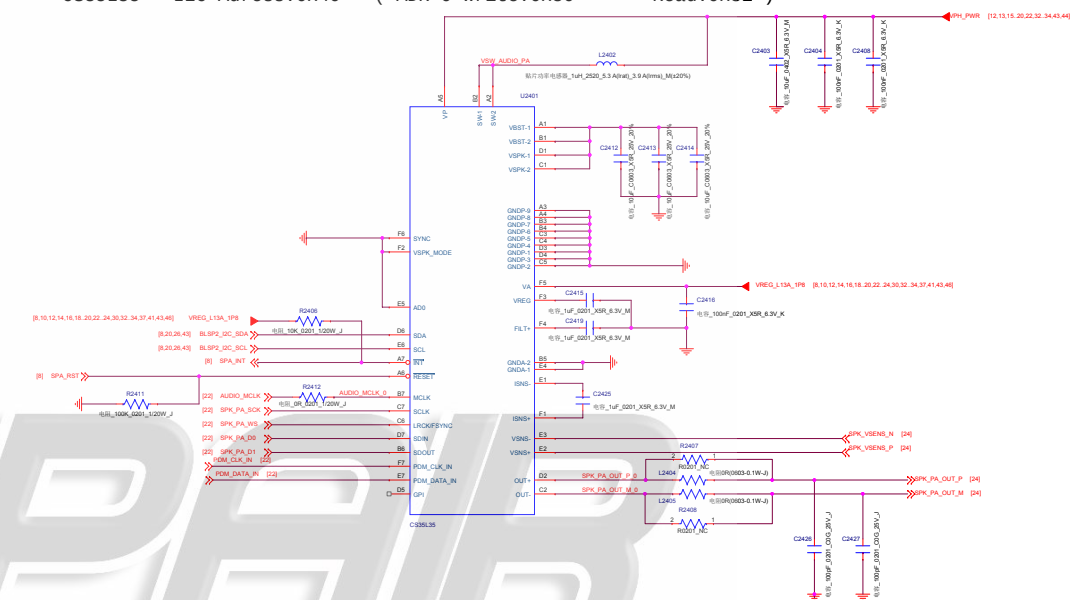
SMART PA

CS35L35

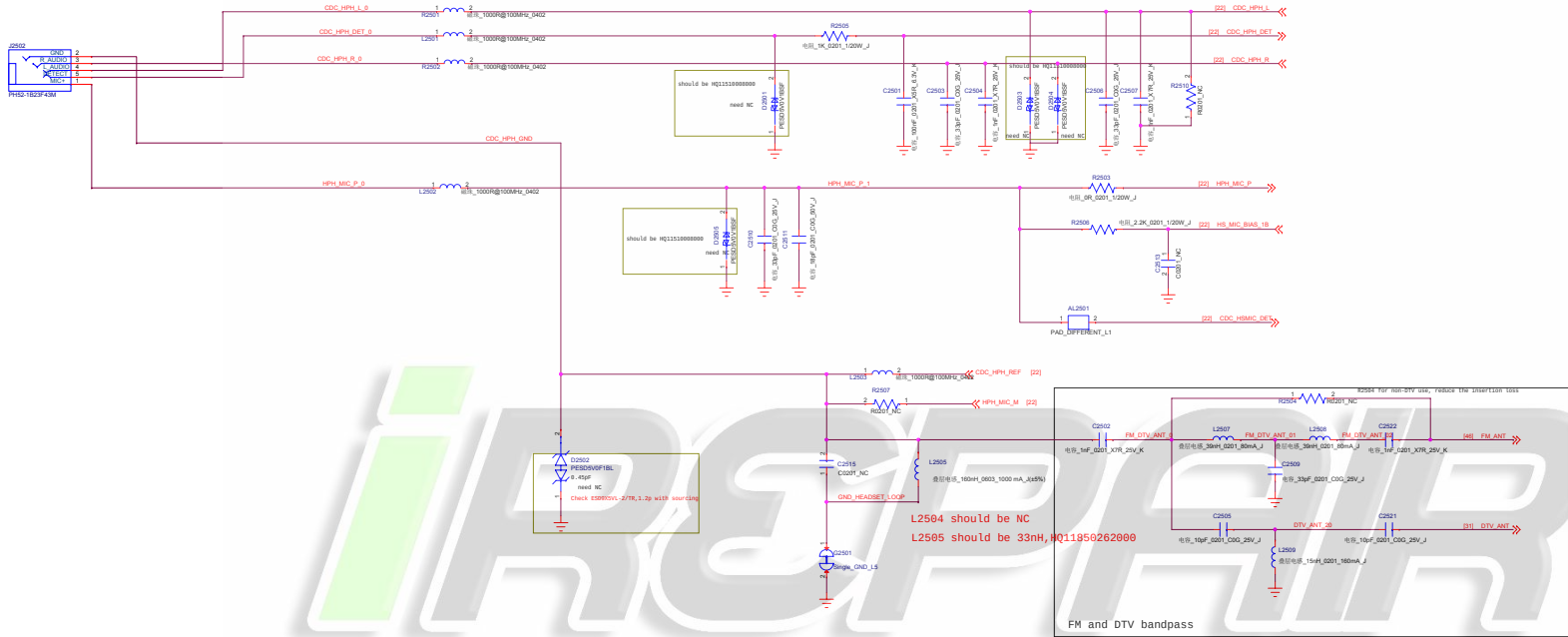
I2C Adress:0x40

```
( ADR=0 Write:0x80
```

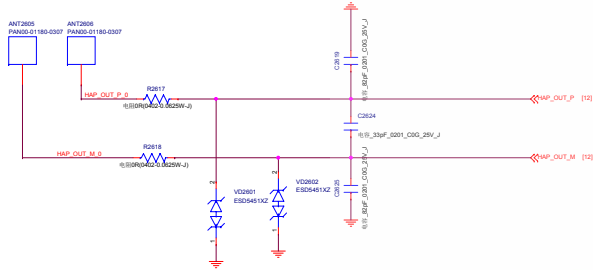
Read:0x81)



AUDIO JACK

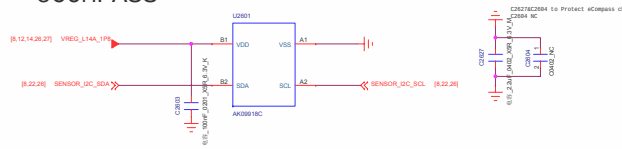


MOTOR



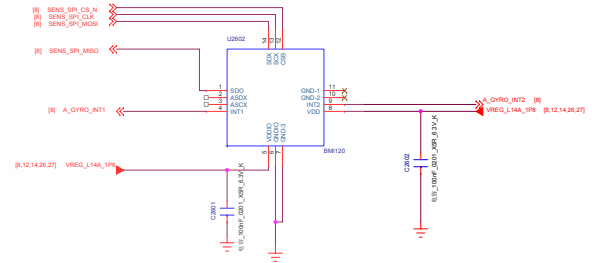
eCOMPASS AK89918C I2C Address:0x0C (Write:0x18

Read:0x19)



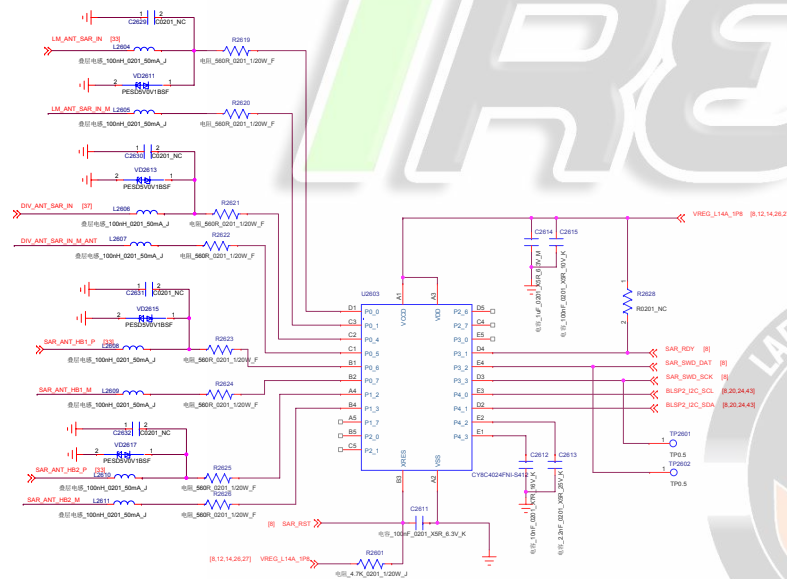
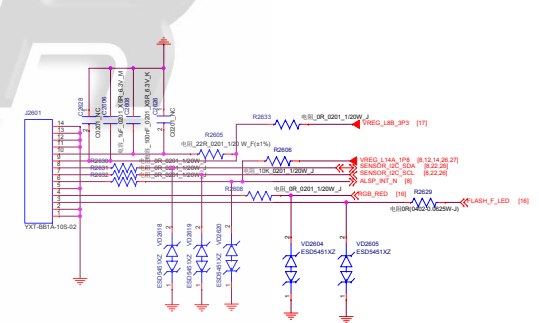
Gyro+G-sensor

BMI120 SPI interface



FPC: 1 ALS&PS, 2 SOL LED, 3 Front Flash LED

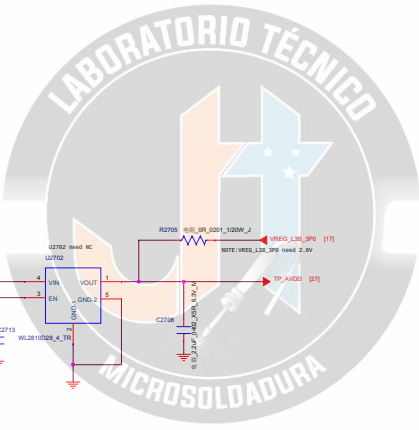
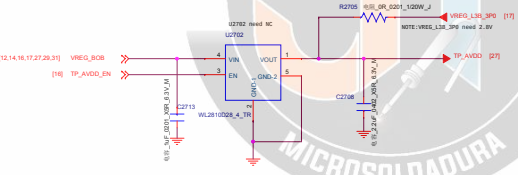
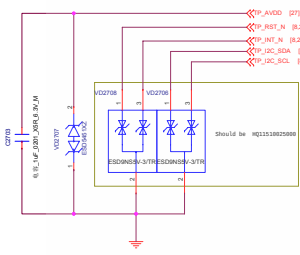
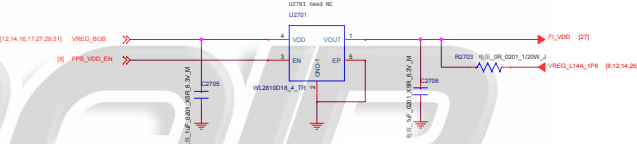
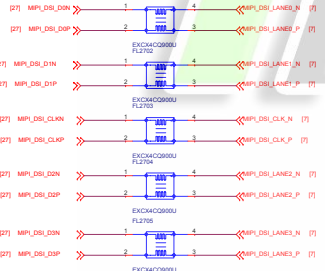
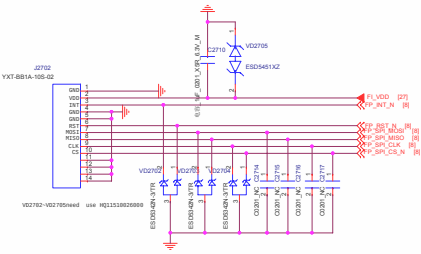
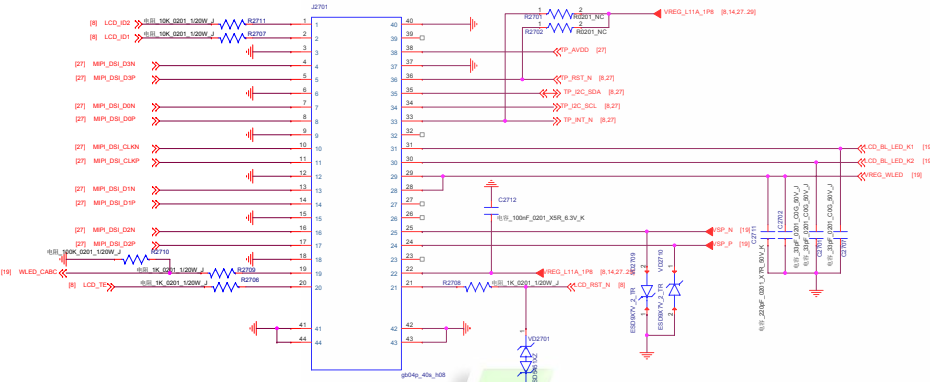
ALS&P:TMD27253 I2C Address:0x39 (Write:0x72 Read:0x73)



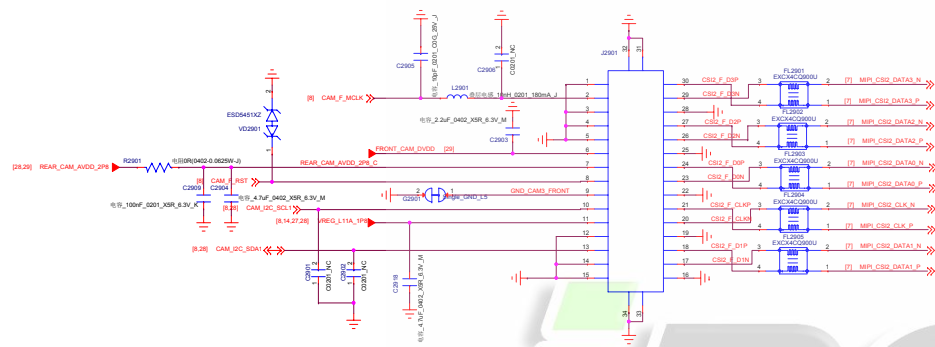
Company Hsuan	
SENSORS_MOTOR	
Taller	1.0
Revisión: 2601-2699	26

TP GT917D I2C Address:0x5D (Write:0xBA Read:0xBB)
TP FT5446 I2C Address:0x38 (Write:0x70 Read:0x71)

FPS Interface and Power



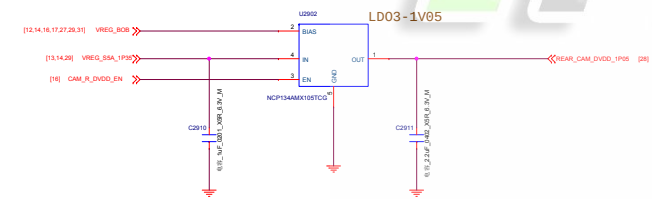
FRONT CAMERA
S5K4H7YX-Front Cam I2C Address:0x10 (Write:0x20 Read:0x21)



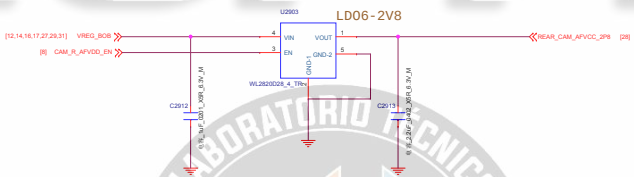
Camera Power Supply Map

	Front CAM	REAR CAM_MAIN	REAR CAM_SLAVE
DVDD	8M - - LD01-1V2 12/20 M - -LD02-1V	LD03-1V05	LD01-1V2
AVDD	LD04-2V8		LD05-2V8
IOVDD	VREG_L11A		
AFVDD	/	LD06-2V8 or VREG_L8B	/

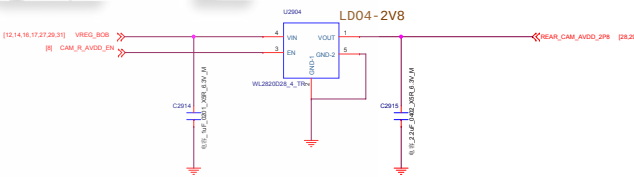
Rear-Main DVDD 1.05V



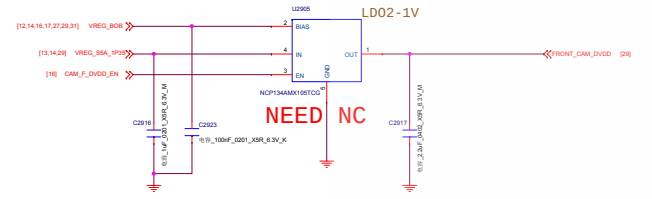
Rear-Main AFVCC 2.8V



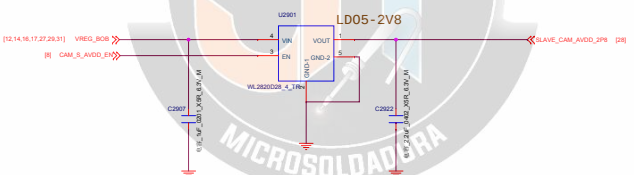
Rear-Main AVDD 2.8V (also Front AVDD)



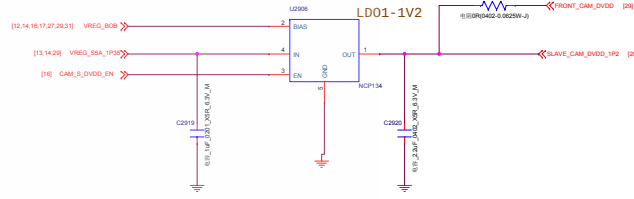
Front DVDD 1V for Front 16M

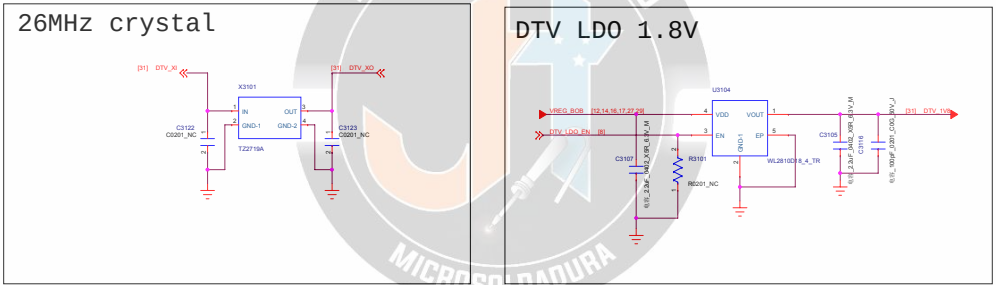
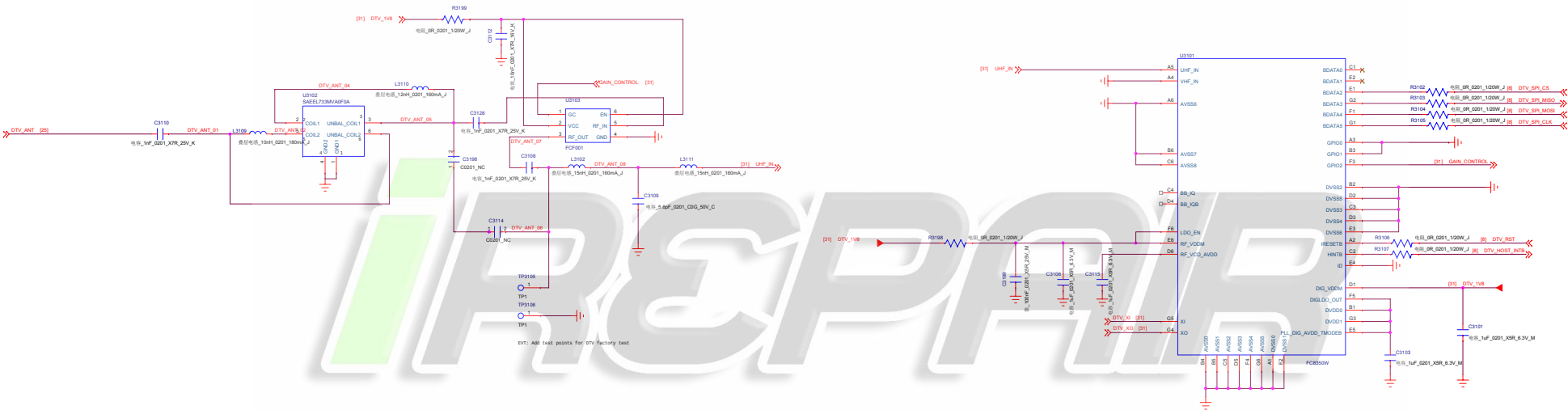


Rear-Slave AVDD 2.8V

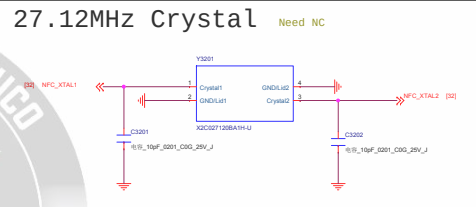
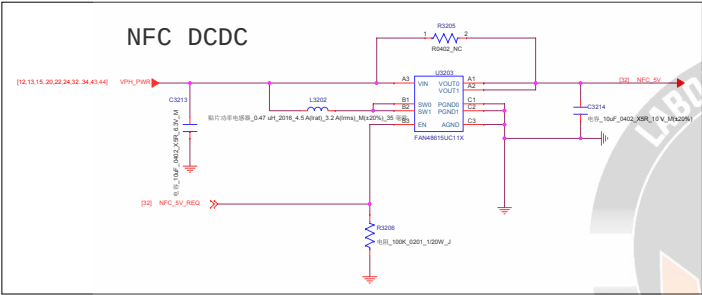
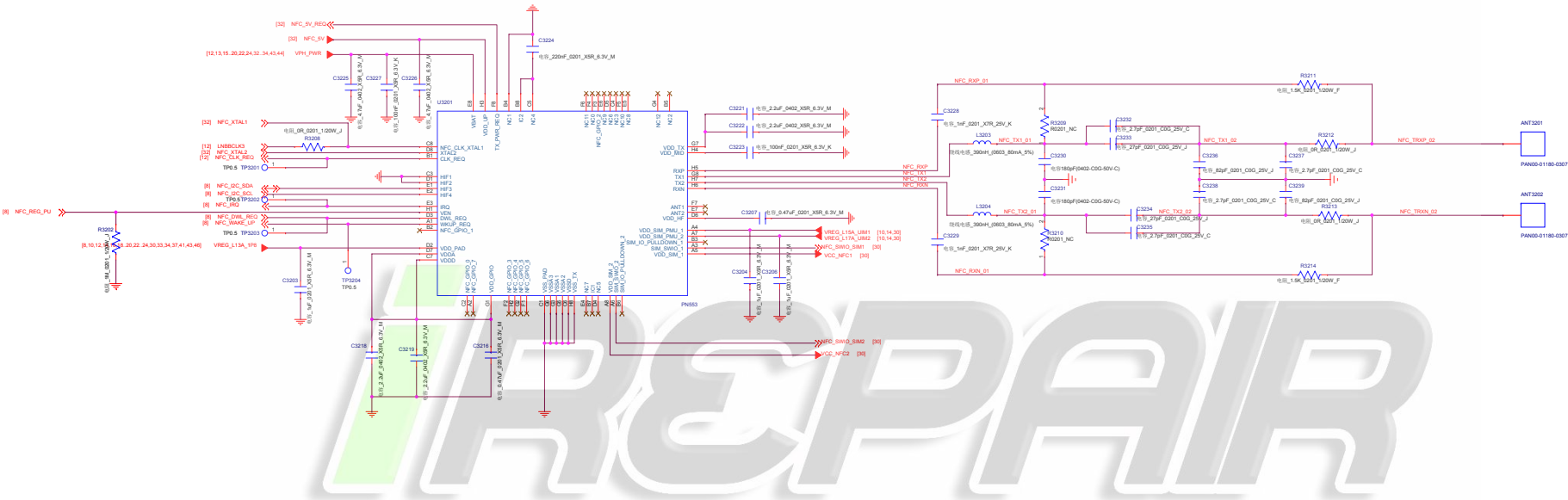


Rear-Slave DVDD 1.2V (Also 8M Front DVDD)





PN553 I2C Address:0x28 (Write:0x50 Read:0x51)



Antenna Tuner

ASDIV_DPDT

DIV_HB

Main_HB

ANT HB R

ANT HB L

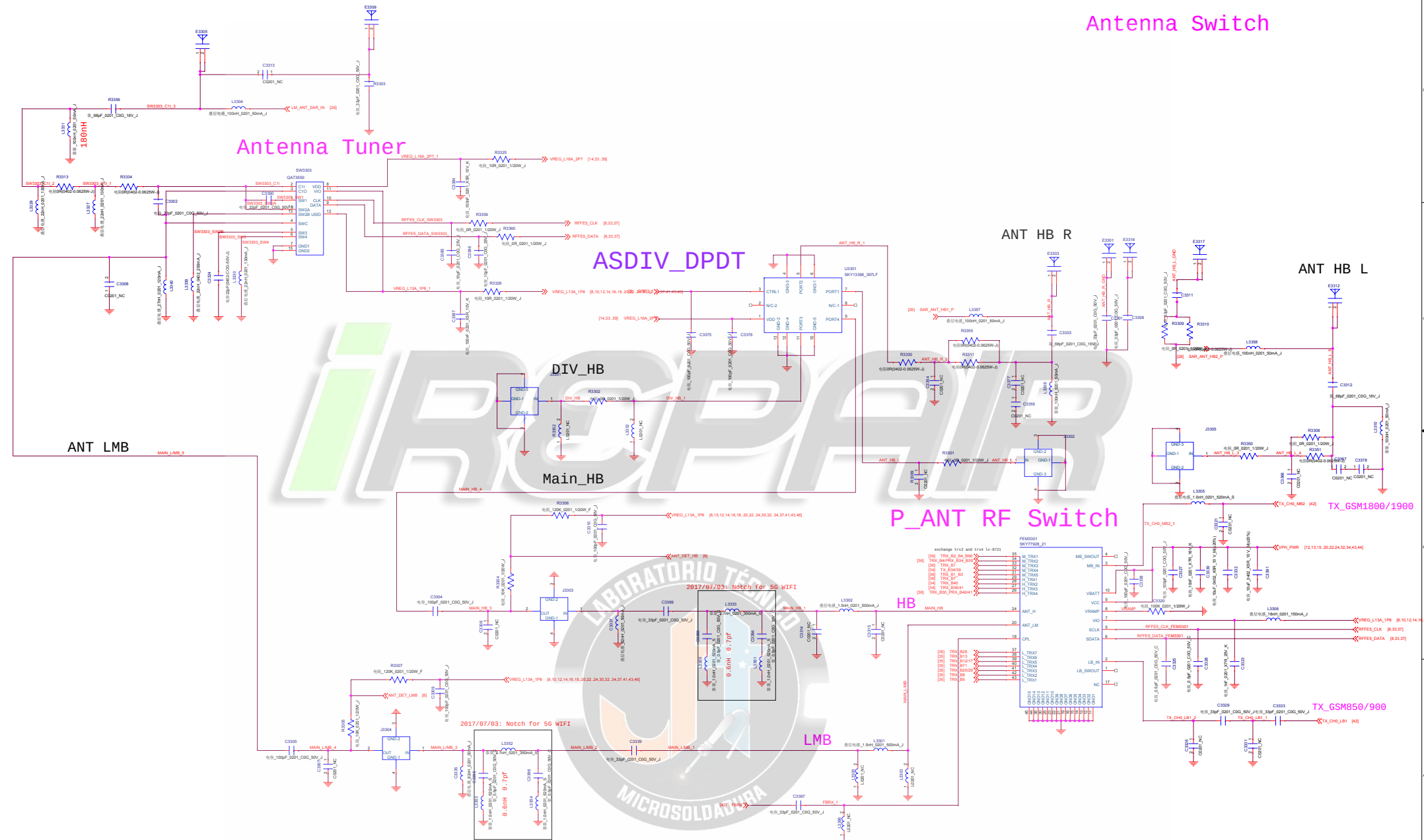
P_ANT RF Switch

HB

LMB

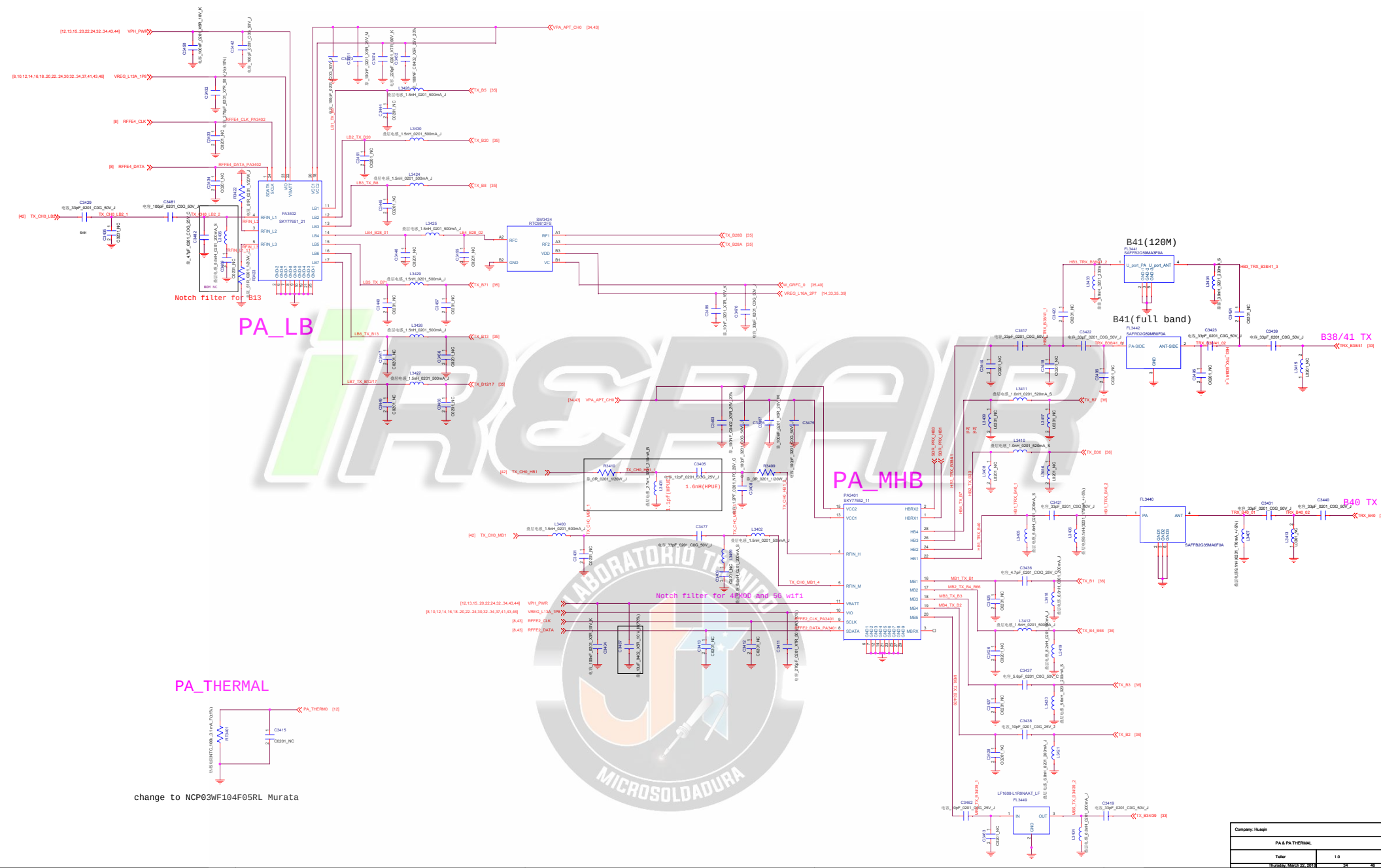
TX_GSM1800/1900

TX_GSM850/900

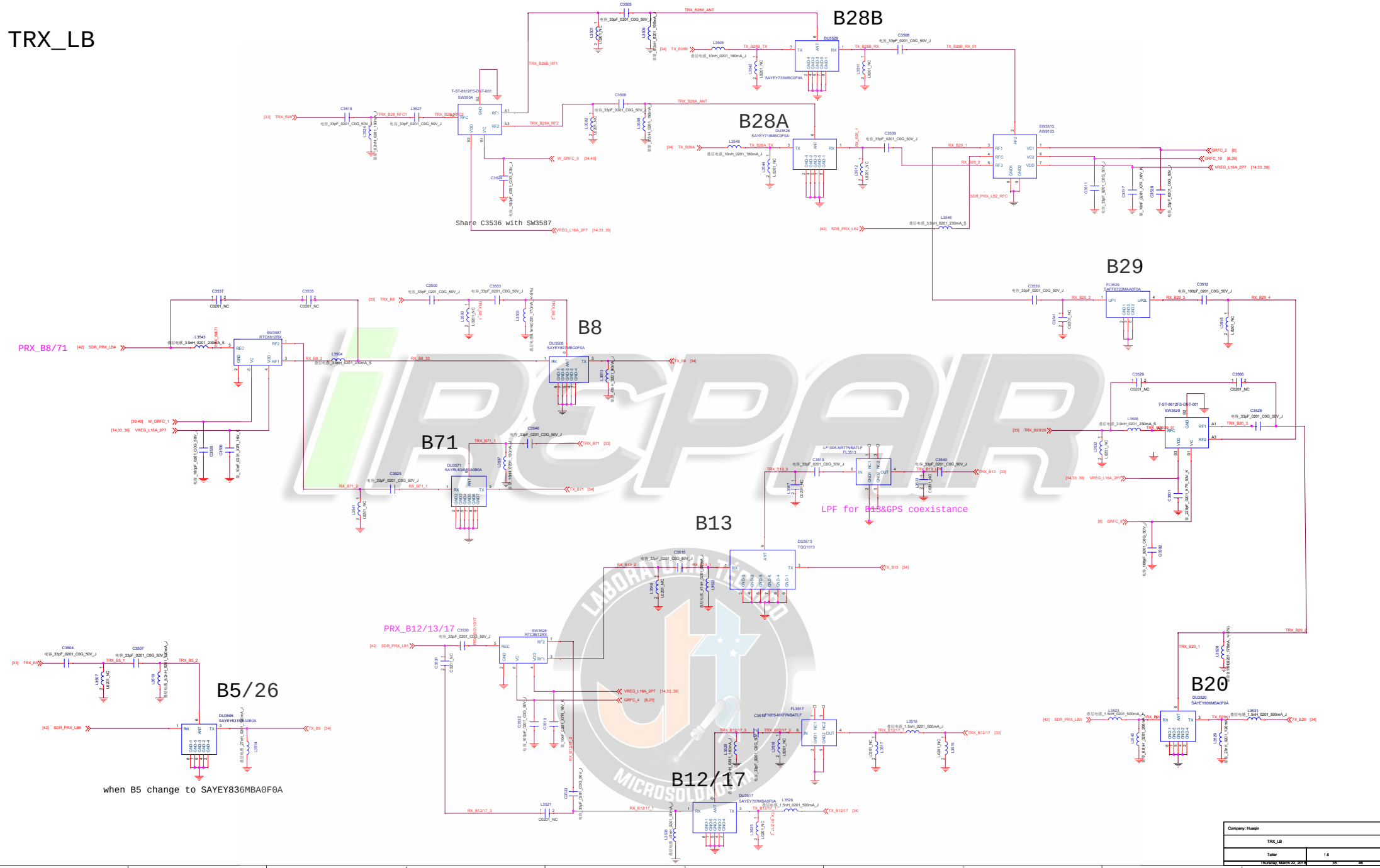


PA & PA THERMAL

REF 3401-3499



TRX_LB



B2 DPX

B34/39 PRx

B2/25 DPX
B4/66 DPX

B25/B66 QPX

PRX_B4

GPIO97

B30 DPX

B3 DPX

B1/B3 QPX

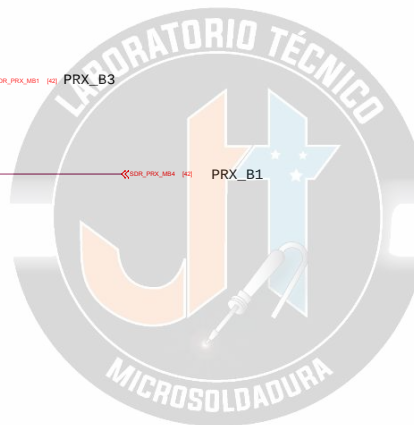
B3 TX
B1 TX

PRX_B3

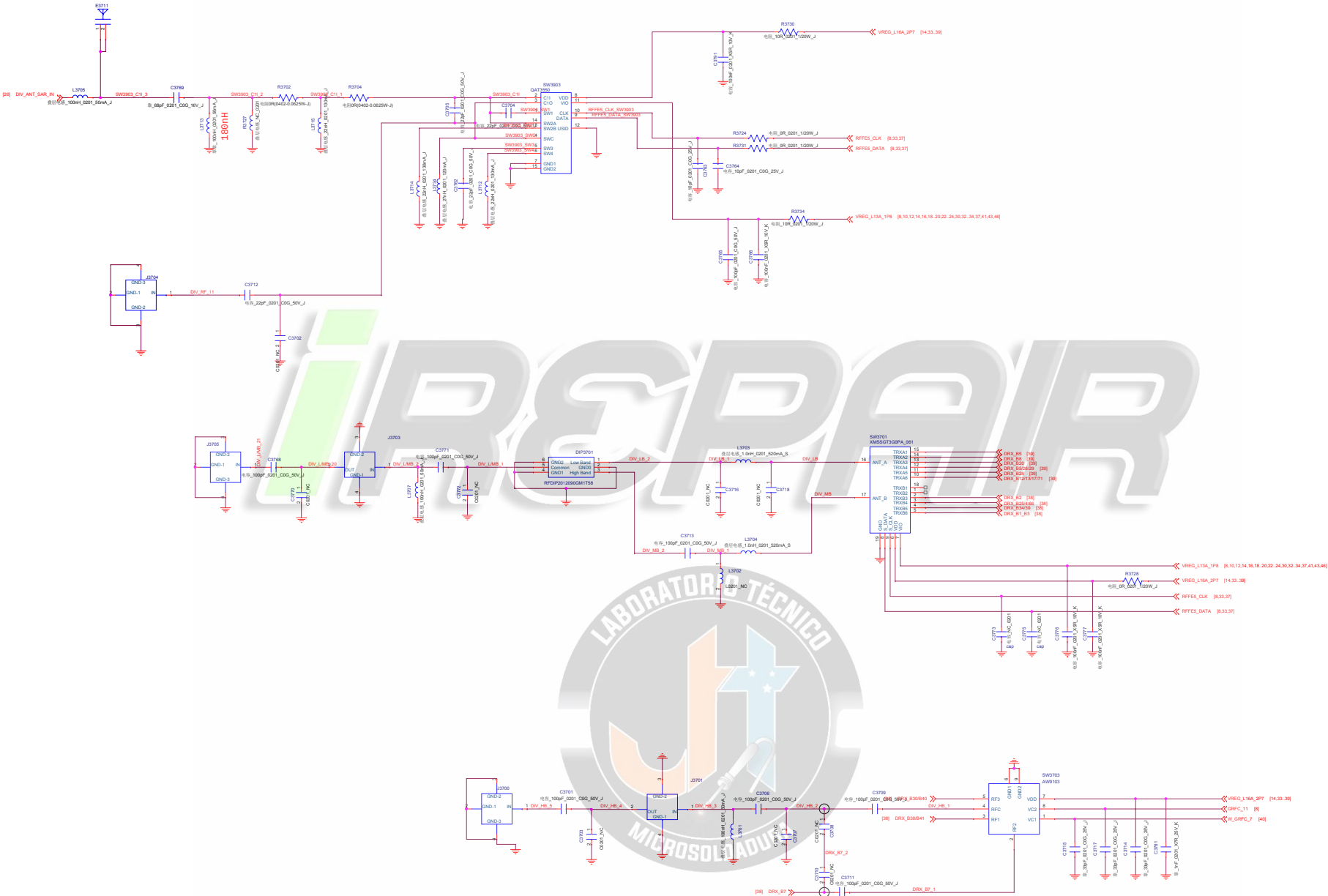
PRX_B1

B7 DPX

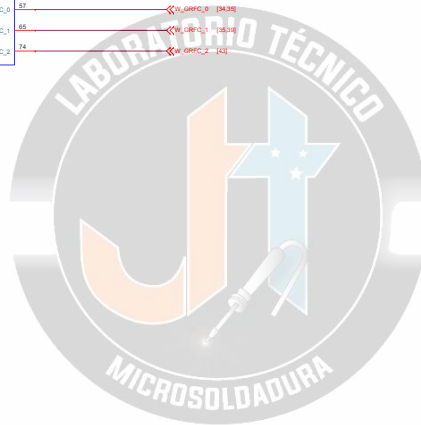
B1 DPX

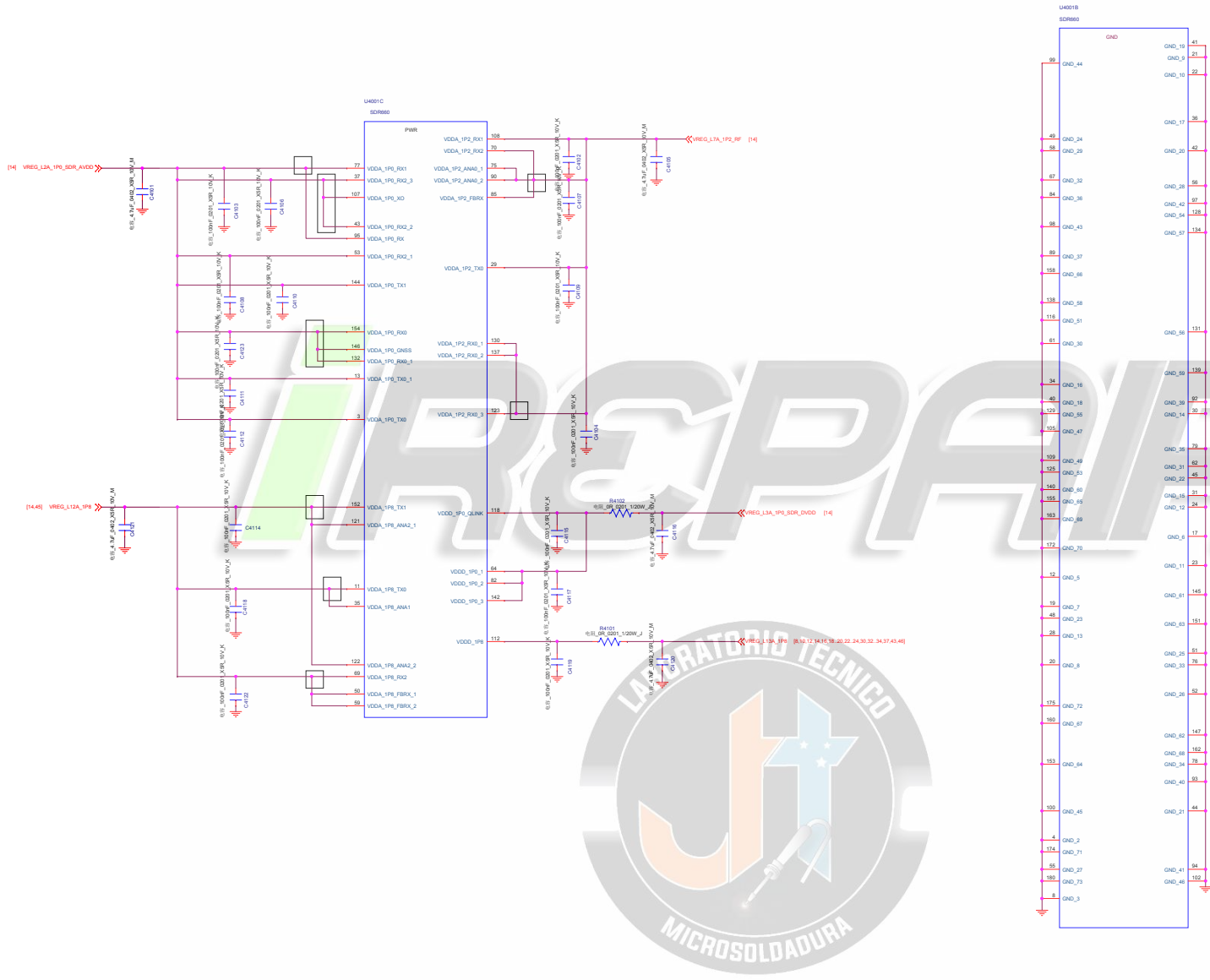


DIVERSIT_ANT

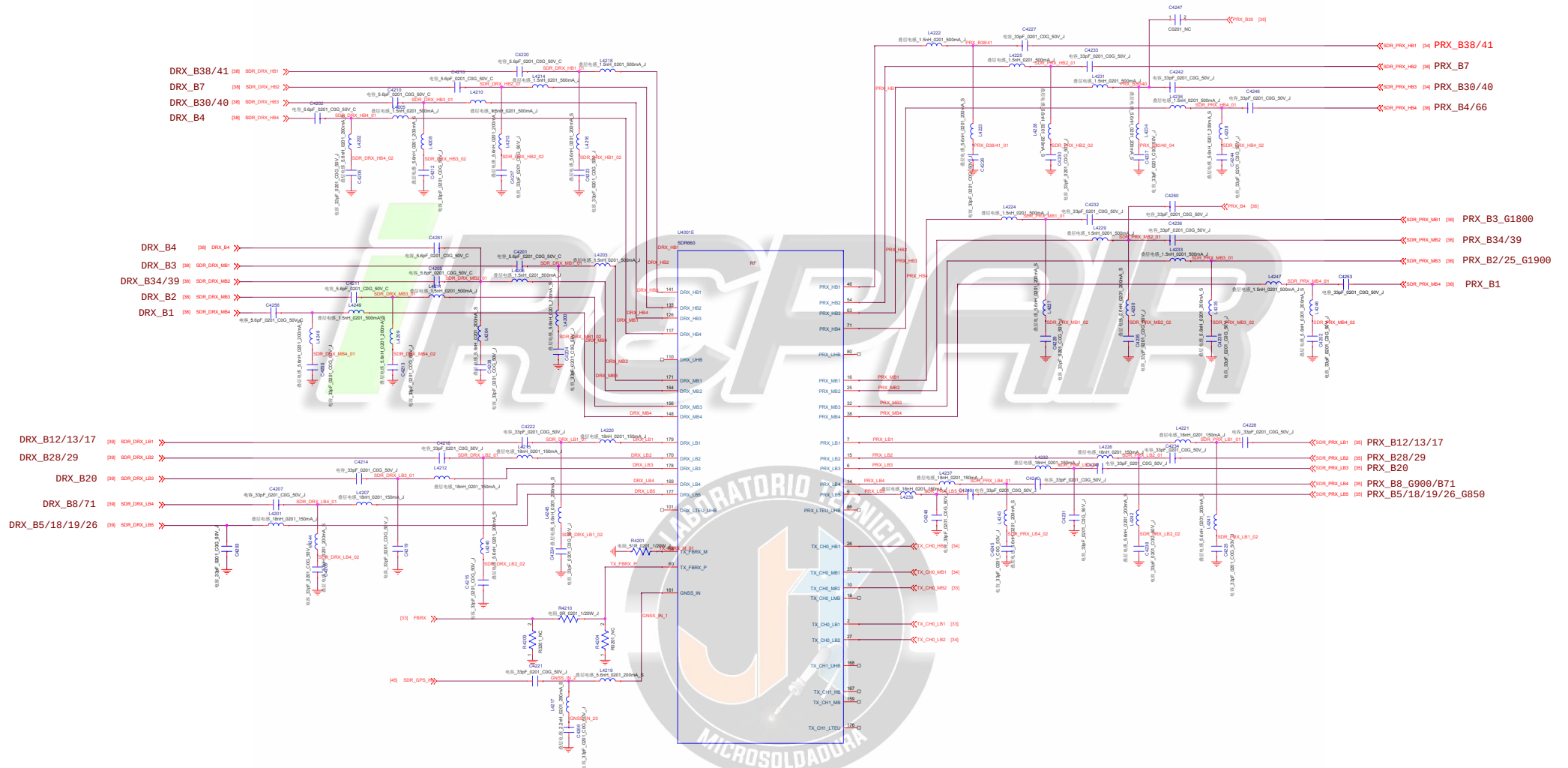


Company Huang	
DIVERSIT_ANT	
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Revisión: 02/08/2018	37

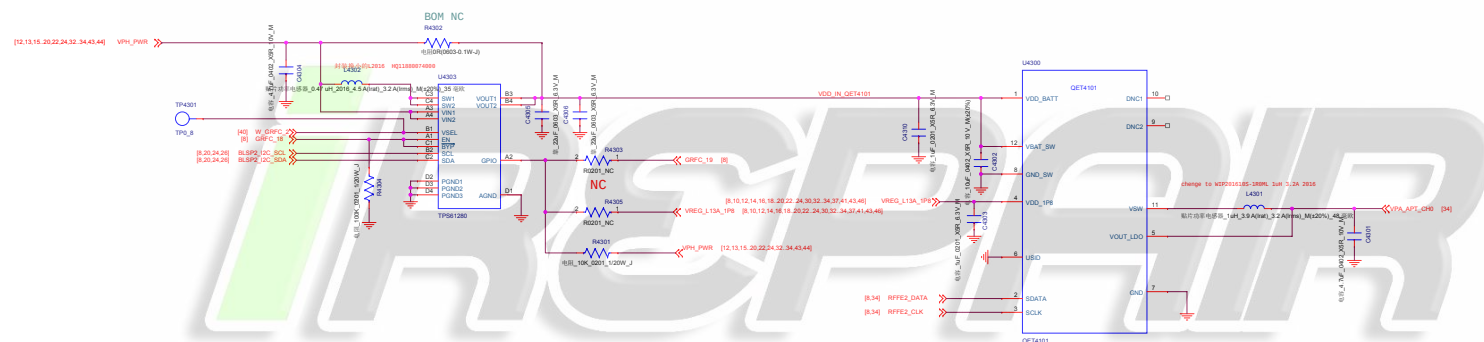




SDR660_DA/PRX/DRX/FBRX



QET4101 & HPUE Boost



WIFI 2.4G/5G/BT

